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Development and Application of an End-of-Life Framework

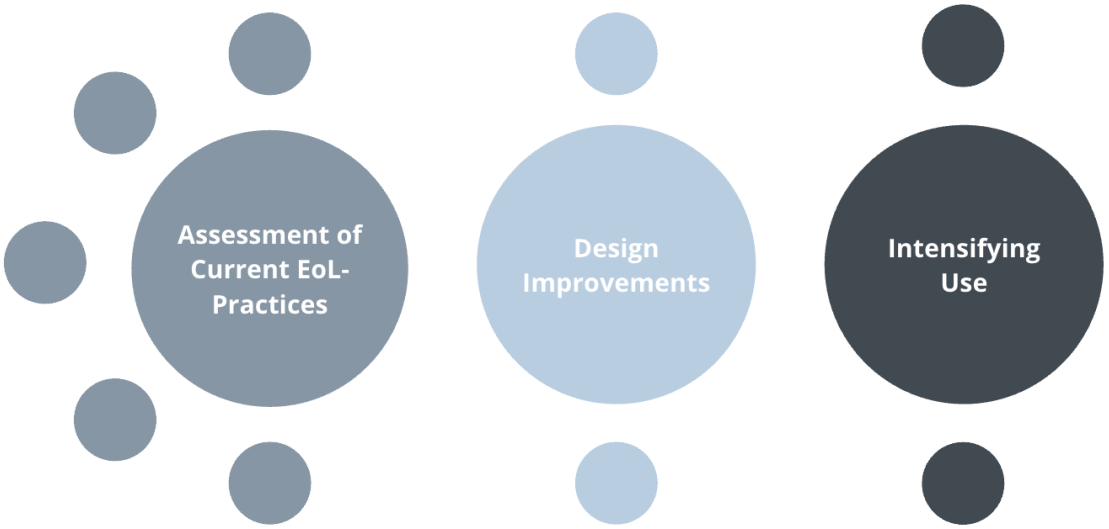
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Development and Application of an End-of-Life Framework

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Utveckling och applikation av ett ramverk för hantering av produkter i deras slutfas

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Sammanfattning

I takt med att den globala klimatkrisen blir mer akut tvingas organisationer hitta sätt att minska sitt miljöavtryck för att möta skiftande klimatpolitik och konsumentkrav. När fordonsindustrin anpassar sig till övergången från förbränningsmotorer till elektrifiering, såväl som övergången från traditionella linjära affärsmodeller till cirkulära, finns det en möjlighet för nya aktörer som kommer in på marknaden att ta till sig cirkulära affärsmodeller från start, vilket ger dem ett försprång gentemot etablerade företag som har långsammare anpassningstider. Cake 0 Emission (Allmänt känt som “CAKE”) är en av dessa nya aktörer, som har visionen att skapa eldrivna motorcyklar med minimal miljöpåverkan. Eftersom deras motorcyklar ännu inte har nått slutet av deras produktlivscykler, har företaget ett behov av att undersöka hur de kan hanteras på ett ansvarsfullt sätt i slutfasen.

Ett ramverk i tre steg har därför skapats för att underlätta hanteringen av produkter i slutfasen för tillverkare av eldrivna motorcyklar. De tre inkluderade aspekterna är: 1. bedömning av nuvarande förutsättningar för slutfashantering, 2. hur produkten kan förbättras genom materialval och designförbättringar, och 3. hur affärsmodellen kan intensifieras för att minska miljöpåverkan av värdeerbjudandet. Det skapade ramverket implementerades sedan på CAKE:s business-to-business motorcykel Makka:work. Produkten analyserades på komponentnivå för att undersöka dess demonteringsnivå, dess återanvändningspotential och potentiella design- och materialförbättringar. Företagets nuvarande affärsmodell undersöktes också översiktligt för att se om förbättringar kunde göras ur ett hållbarhetsperspektiv.

KTH Industrial Engineering and Management

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Abstract

As the global climate crisis becomes more urgent, organizations are forced to find ways of reducing their environmental footprint to meet changing climate policies and consumer demands. As the automotive industry is adapting to the shift from internal combustion engines to electrification, as well as the shift from traditional linear business models to circular ones, there is a chance for new actors entering the market to adopt circular business models from their inception, giving them an edge over incumbent firms who have slower adaption times. Cake 0 Emission (generally referred to as “CAKE”) is one of these new actors, who has the vision of creating electric two-wheelers with minimum environmental impact. As their motorbikes are yet to reach the end of their product life cycles, the company has a need to investigate how they can be properly managed in the end-of-life phase.

A three step framework has therefore been created to facilitate the end-of-life practices for electric motorbike manufacturers. The three included aspects are: 1. the assessment of the current prerequisites for end-of-life management, 2. how it can be improved through material choice and design improvements, and 3. how the business model can be intensified to decrease the environmental impact of the product-service offerings. The created framework was then implemented on one of CAKE’s business-to-business motorbikes called Makka:work, where the product was analyzed on a component level to investigate its level of disassembly, its second life potential and potential design and material improvements. The company’s current business model was also broadly investigated to see whether improvements could be made from a sustainability perspective.

Foreword

This Master's thesis has been fulfilled as a part of the master´s programme Innovation Management and Product Development at KTH Royal Institute of Technology in Stockholm. The project was performed in collaboration with the sustainability department at Cake 0 Emission.

We would like to thank our supervisors at CAKE, Isabella Pehrsson and Avinash Kumar, for their support, inputs, and guidance throughout the project. Additionally, we would like to acknowledge how valuable the inputs from the interview respondents have been in this project. Lastly, we want to express our gratitude to our supervisor at KTH Royal Institute of Technology, Johan Arekrans, for giving us support throughout the master thesis project.

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Abbreviations

B2B	Business-to-Business
CBA	Cost-Benefit Analysis
CE	Circular Economy
DfD	Design for Disassembly
DfE	Design for Environment
DfM	Design for Modularity
DfR	Design for Recycling
DfX	Design for X
EoL	End-of-Life
ELV	End-of-Life Vehicle
EV	Electric Vehicle
EVB	Electric Vehicle Battery
EU	European Union
ICE	Internal Combustion Engine
ICEV	Internal Combustion Engine Vehicle
IoT	Internet of Things
LCA	Life Cycle Assessment
LCC	Life Cycle Cost
LIB	Lithium-Ion Battery
PSS	Product-Service Systems
RL	Reverse Logistics
RSC	Reverse Supply Chains

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1. Introduction

This chapter gives an introduction to the project by describing the background of the research area, the purpose and delimitations of the study.

1.1 Background

Sustainability has become a pressing issue in the automotive industry. As the global climate crisis becomes more urgent, most industries are forced to find ways to reduce emissions and decrease their environmental impact. In recent years, the need for shifting from a linear to a circular economy has become heavily emphasized, where the traditional linear model has a sole purpose of profitability irrespective of the impact the product life cycles have on the environment (Hofman, 2019). In a circular model, companies have to account for the entire life cycles of their products, from extraction of raw materials to the end-of-life (EoL) phase. The public awareness regarding climate issues is at an all-time high, causing a consumer demand for organizations to become more sustainable. In addition, new policies are constantly emerging on both national and European Union (EU)-levels to decrease the rate in which greenhouse gases are emitted and finite natural resources are exploited. One industry that has been subjected to changed requirements due to regulations is the automotive industry. Vehicles in the end of their lifespan have not been properly managed in the past resulting in large waste streams and environmental problems (Chaabane et al., 2021; European Commission, 2000). Because of this, the EU has listed several requirements for end-of-life vehicles (ELVs) in Law/Directive 2000/53/EC, which requires companies to factor in dismantling, reuse and recovery of vehicles in the product design, to have a collection system of ELVs, to not use hazardous substances among other aspects. Even though these requirements are specified for passenger cars and small trucks as of today, the same factors must be considered by manufacturers of all types of vehicles if they have aspirations of being sustainable.

The automotive industry has faced another major disruption with the shift from internal combustion engines (ICE) driven by fossil fuels to the introduction of electrification. When disruptions occur in an industry, incumbent firms must make major changes to their deliveries. The biggest actors often have the more rigid systems when it comes to quickly adapting to market changes, and most often also have the highest risk involved in rapidly changing their value propositions. This opens up avenues for new actors to enter the market as they are not held back by the equivalent firm rigidities and value lock-ins (Hölttä-Otto et al., 2013). Cake 0 Emission (Generally referred to as “CAKE”) is one of these new actors, who have identified a gap on the market for high performing electric motorbikes, and have been rapidly growing since their founding in 2016. Their vision is to make motorbikes with as close to zero-emissions as possible, while following their mantra “light, quiet and clean”. As they are establishing themselves, they are also investigating ways of optimizing their product lifecycles from a sustainability perspective. Since their first motorbike was released onto the market in 2018, none of their products have naturally reached their end-of-life phase

as the technical lifespan of the bikes is estimated to be 20 years or a driving distance of 600 000 km. Therefore, there is a need to establish how the bikes should be managed once they reach this phase of the lifecycle. Additionally, the company has a need to investigate suitable circular business models to promote sustainable end-of-life practices.

1.2 Purpose

This master thesis project focuses on circular business models to help make electric two-wheelers more sustainable in the end-of-life phase. Based on an analysis of their current strategies, the purpose of this thesis is to provide guidelines to Cake 0 Emission regarding technical and socio-technical changes in their business model to improve their end-of-life practices. The analysis will focus on one of CAKE's Business-to-Business (B2B) bikes, the electric two-wheeler "Makka:work" and the possible end-of-life strategies for the bike's ingoing components when the bike's lifespan is determined to be 20 years or a travel distance of 600 000 km. However, the guidelines established in the thesis can act as a framework for all of CAKE's electric two-wheelers as they follow similar design principles.

The main objectives of this thesis are to:

- Investigate circular business model deliveries and select key aspects to be included in an EoL-framework.
- Analyze possible EoL-options by selecting suitable EoL-strategies for all ingoing components of Makka:work.
- Give suggestions on how Cake 0 Emission can adapt their business model to improve their end-of-life practices.

1.3 Delimitations

While the analysis covers an electric two-wheeler, it is not centered around the batteries in the company's products as Cake 0 Emission already investigates specific second life applications for the batteries used in their bikes. Therefore, the analysis has been made to include options for all other ingoing parts. The EoL-strategies analyzed for each part have been limited to what can be done internally at CAKE, and thus, figuring out how external actors could repurpose certain parts of the bike is outside of the thesis scope. In addition, the thesis has mainly focused on environmental sustainability deliveries. The economic viability of certain EoL-strategies has been assessed in a general sense, but no cost estimation methods have been used to analyze the cost efficiency of end-of-life choices. Similarly, the aspect of social sustainability has also been deprioritized to put a larger emphasis on the environmental sustainability aspect of EoL. Further, the potential EoL-options in this thesis are based on the possibilities on the Swedish market, as the prerequisites for CAKE to implement the same EoL-strategies are not as established internationally. Lastly, due to limited information regarding the weight of different mechanical fasteners used in the case vehicle, 205 mechanical fasteners have been assumed to weigh 5 grams each.

2. Frame of Reference

In this chapter, previously written literature and conducted research for end-of-life management is presented. The included topics are based on what are seen to be key aspects of end-of-life management for electric two–wheelers. The frame of reference covers the definition of an ELV, legislations and policies regarding ELVs, and different suggestions on how to manage vehicles when they have reached the EoL-stage (referred to as *R-strategies*). Moreover, circular business models, the environmental challenges with managing electric vehicles (EVs), and different design strategies to improve EoL-management are also covered in this chapter.

2.1 End-of-Life Vehicle Definition

Go et al. (2011) define an ELV as a vehicle which is to be discarded by its registered owner as waste. They have the potential of emitting a vast amount of pollution if they are not managed properly (Karagoz et al., 2019). Vehicles are seen to have different use phase lengths depending on whether the country of its use is developed, developing or underdeveloped. In developed countries, the general vehicle use phase is said to be 10-12 years (Mayyas et al., 2012). In developing and underdeveloped countries, the use phase is said to be around 15-20 and 20-25 years respectively (Mayyas et al., 2012). Vehicles typically reach the end-of-life phase due to wear and performance degression caused by age, or by heavy damage due to accidents or other external factors. Consequently, ELVs can either be categorized as natural or premature, where natural ELVs are those who need to be discarded due to age, and premature ones are those who need to be discarded before their intended lifespan has come to an end due to accidents, thefts, fires, vandalism or other external unforeseen incidents. Premature ELVs often contain parts that are in good condition for reuse as most components have not deteriorated because of age, while parts in natural ELVs often have low reuse and resale value due to wear, safety risks and low repairability and refurbishment potential (Go et al., 2011).

2.2 End-of-Life Vehicle Legislation

As time goes by, more sectors are becoming regulated to halt the environmental problems connected to over-exploitation of finite resources, emission of greenhouse gasses and insufficient waste management. According to the World Population Prospects data by the United Nations, the world's population grew from 2.5 billion to 7.6 billion people between 1950 and 2018, an annual growth rate of 1.6 percent (Milani et al., 2021). With how rapidly the earth population is growing, along with the technological and economical advancements of developing countries, the demand for private transportation is growing at an unprecedented rate placing a larger emphasis on establishing an environmentally sustainable vehicle industry. Therefore, policies focused on reducing the environmental burden of ELVs have been put in place to increase the producer responsibility in the private car industry (European Commission, 2000), and further policies will likely continue to emerge.

2.2.1 European Union Directive

The European Union has released Law/Directive 2000/53/EC with the aim of preventing and limiting waste from ELVs and their parts by ensuring their reuse, recycling and recovery (European Commission, 2000). It also aims to improve the environmental performance of all the involved stakeholders of the vehicle lifecycle (ibid.). Manufacturers must factor in disassembly, reuse, recycling and recovery of vehicles in the design and production phase to ensure that they are reusable and/or recyclable to a minimum of 85 % of the vehicles total weight, and reusable and/or recoverable to a minimum of 95% of its weight (European Commission, 2000). They are also restricted in the use of certain hazardous substances such as lead, mercury among others (ibid.). As of today, the directive is specified for passenger cars and small trucks but there are ongoing discussions of including motorcycles in the future (FEMA, 2022). The inclusion of motorcycles in the directive would mean that new guidelines would be in place for motorcycles being scrapped in Europe (Henshaw, 2022; European Commission, 2000), leading to larger producer responsibility when it comes to end-of-life management of two-wheelers.

2.2.2 Reverse Logistics

Generally, when a product has been sold, the responsibility of how it is discarded by the end of its lifespan is shifted from the seller to the customer. One of the requirements for sustainable end-of-life management, which is also included in Law/Directive 2000/53/EC, is that manufacturers, importers and distributors must provide systems to collect ELVs and use their parts where it is technically feasible (European Commission, 2000). For the parts of these ELVs to be responsibly managed by its parent company, there is a need for them to be returned from the consumer, an area identified in literature as reverse logistics (RL). In many cases, this methodology can be used interchangeably with the term reverse supply chains (RSC) (Ene & Özturk, 2015).

Ene & Özturk (2015) describe reverse flows as the collection of used products from consumers and the transportation of these products for the purpose of recycling, reuse or disposal. The aim is to avoid the prospect of them ending up in unwanted waste streams. Vehicles can be returned According to Beullens et al. (2004), a functioning collection system requires the users to participate consistently in the reverse logistics of the ELV. The user behavior can be stimulated by having collection programs which deliver good service in terms of consistency and being on time. Additionally, the collection and transport system must be designed in view of the targeted reprocessing application. Reuse requires the ELV to be returned in the best condition possible, while material recovery is less sensitive to damaged or worn goods (ibid.). There is also a need of having the ELVs transported in a cost-efficient way to the sites of the RL network, which may call for temporary storage and accumulation of ELVs before being shipped. Other aspects such as volume compaction, source separation and vehicle characteristics may have to be taken into consideration to effectivize the collection and transport system (ibid.).

2.3 R-Strategies

The r-strategies, sometimes known as the 10 Rs of Circular Economy, are the different methods of making a product lifecycle more environmentally sustainable. The different r-strategies are used to minimize the amount of waste that has been shown to emerge in the traditional linear economy models to create higher resource efficiency (Morseletto, 2020). The ways of doing so are by using and manufacturing products in a smarter way, extending the product life cycles, and by useful application of materials (Potting et al., 2017). The r-strategies are further categorized into these different ways of decreasing waste and improving resource efficiency, as can be seen in Table 1 below.

Table 1: R-strategies, based on Potting et al. (2017).

Smarter product use and manufacture	R0	Refuse	Make product redundant by abandoning its function or by offering the same function with a radically different product
	R1	Rethink	Make product use more intensive (e.g. through sharing products or by putting multi-functional products on market)
	R2	Reduce	Increase efficiency in product manufacture or use by consuming fewer natural resources
Extend lifespan of product and its parts	R3	Reuse	Re-use by another consumer of discarded product which is still in good condition and fulfils its original function
	R4	Repair	Repair and maintenance of defective product so it can be used with its original function
	R5	Refurbish	Restore an old product and bring it up to date
	R6	Remanufacture	Use parts of discarded product in a new product with the same function
	R7	Repurpose	Use discarded products or its part in a new product with a different function
Useful application of materials	R8	Recycle	Process materials to obtain the same (high grade) or lower (low grade) quality
	R9	Recovery	Incineration of material with energy recovery

There are several important factors to consider when assessing which r-strategy should be selected for product components. According to Hedayati & Subic (2011), sustainable handling of ELVs consider environmental, economic and social aspects. The authors also emphasize that the options differ depending on which country is carrying out the ELV practices, where EU countries must have a 95% recovery rate while the legislation in other countries do not require the same recovery rate (Hedayati & Subic, 2011). The techniques and general acceptance for ELV recovery is consequently varied in different countries.

As seen in *Table 1*, a majority of the r-strategies have the purpose of extending the lifespan of products and their parts. It is important to not only consider how the entire vehicle can be scrapped in the most environmental way, but rather how the product life span can be prolonged through continuous use of the r-strategies *reuse, repair, refurbish, remanufacture*

and *repurpose*. According to Korhonen et al. (2018), reuse, remanufacturing, and refurbishing demands less energy and fewer resources than traditional recycling, resulting in economic and environmental benefits for manufacturers to implement these strategies instead. To elaborate, recycling aims to circulate material used in products while r-strategies *R3-R7* (shown in *Table 1*) aim to prolong the life of a product or a component (Alamerew & Brissaud, 2019). Consequently, recycling is seen as less desirable than life extension strategies as they do not require the same resources in terms of sorting, shredding/melting and reprocessing materials (ibid.). This notion is further consolidated by the waste management hierarchy model (see *Figure 1* below), where the environmental impact of every EoL-strategy is categorized from most to least preferred.

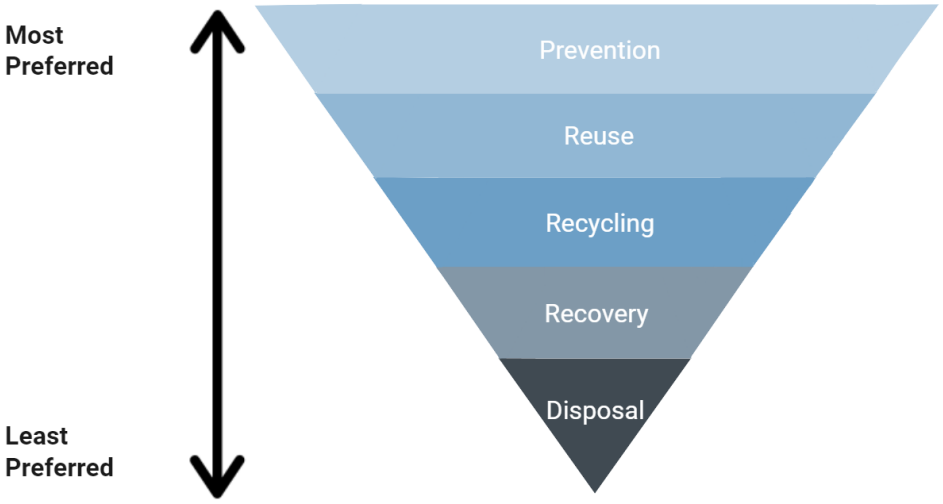


Figure 1: Waste management hierarchy model, based on European Commission (n.d.-a).

Bakker et al. (2014) deliver descriptions of the different categories included in the waste management hierarchy, where *prevention* is described as "Measures taken to reduce the quantity of waste". This includes all the strategies used to reduce wasteful material utilization in products. The category *reuse* is described as "Using a product or component again for the same purpose for which it was conceived". This includes all the strategies where products or product parts are reused or reconditioned in some way to be used in a second life cycle. When connecting the different r-strategies to the waste hierarchy model in this thesis work, the r-strategies *R0-R2* are therefore placed under the hierarchical category *prevention*, *R3-R7* under *reuse*, *R8* under *recycling* and *R9* under *recovery*.

Furthermore, there are several evaluation methods for choosing r-strategies encountered in literature. Schmid et al. (2016) used a multi-criteria analysis with nine different criteria for analyzing how parts of the vehicle should be recovered, and these were based on different environmental impact categories, cost estimation methods or technical performance metrics. The PROMETHEE-method (Preference Ranking Organization METHod for the Enrichment of Evaluations) was used for evaluating which r-strategies were the best options by comparing them in respect to the defined criteria (Schmid et al., 2016). Okomura et al. (2016) developed a multi-agent system for analyzing recovery options, and these were based on the environmental impact and incurred cost of different EoL-options.

2.4 Circular Business Models

A traditional linear economy model is based on product usage from "cradle-to-grave", meaning a product lifecycle from the extraction of raw materials to when it is discarded by its user as waste (Morseletto, 2023). It is seen as an economy that goes from extracting virgin raw materials which later end up in landfills without any recycling considerations (ibid.). Electric vehicles have the potential of delivering sustainable transport systems. However, for this to become reality, the entire product life cycle needs to be considered. While electrification will reduce emissions during the product's use phase, the production of electric vehicles is still resource intensive leading to other sustainability issues (Richter, 2022). For the lifecycle to become sustainable there needs to be a shift from a linear to a circular economy (CE). Implementations of a circular economy are meant to reduce pollutants in order to meet global climate goals such as the Paris Agreement, Agenda 2030 and other initiatives. The adaptation of the CE approach is estimated to limit material cost savings of hundreds of billions of dollars annually for the EU and result in environmental benefits (Asif et al., 2021). A fundamental aspect in creating a circular economy is resource efficiency, which means using natural resources in a sustainable manner while minimizing impacts on the environment (European Commission, n.d.-b). It involves accounting for repairability to extend the product lifespan, reuse to exploit component functions in a sustainable way, refurbishing/repurposing to find second life applications for parts of a product, and lastly recycling to circulate material loops (Richter, 2022).

The term circular business models combines the concept of circular economy and traditional business model innovation (Geissdoerfer et al., 2020). A traditional business model describes which strategies a company uses to create, deliver value and capture value (Osterwalder & Pigneur, 2010). In a circular business model, the aim is to change the business model components (i.e. value proposition, value delivery, and value capture) through cycling, extending, intensifying and dematerialising material loops. The deliveries of a circular business model can be seen in *Figure 2* below.

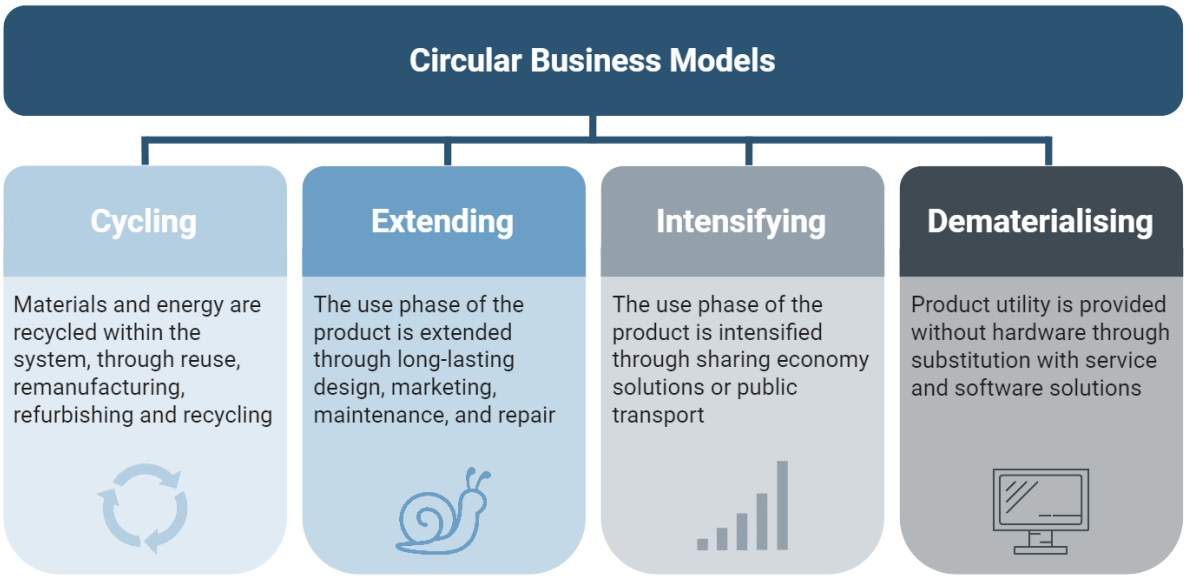


Figure 2: Circular business model deliveries, based on Geissdoerfer et al. (2020).

According to Geiessdoerfer et al. (2020), cycling, extending, intensifying, and dematerialising material and energy loops can be achieved through digitalisation, servitization, sharing solutions, long-lasting product design, maintenance, repair, reuse, remanufacturing, refurbishing, and recycling. For this thesis work, dematerialising is not seen as a main focus as the hardware of a motorbike cannot be substituted through digitalization, and reducing the number of manufactured products by implementing product-service systems (PSS) will be included in the intensifying-delivery. Therefore, the following subsections will encompass the first three deliveries of the circular business model, namely *cycling, extending* and *intensifying.*

2.4.1 Cycling

Cycling is defined by Geissdoerfer et al. (2020) as the recycling of materials and energy within the system through reuse, remanufacturing, refurbishing and recycling. This implies closing material loops by finding second life applications for products and their parts, and thereby decreasing the need for raw material extraction, processing of virgin materials and manufacturing of new parts. This can also be defined as establishing closed resource loops or a cradle-to-cradle lifecycle view, which has two main objectives: (1) reuse of material, and (2) recycling of materials (Bocken et al., 2016). This circular business model objective corresponds to r-strategies R3-R8 previously seen in *Table 1.*

2.4.2 Extending

Geissdoerfer et al. (2020) define extending as the extension of the use phase through long-lasting and timeless design, marketing that encourages long use phases, maintenance and repair. In the study conducted by Potting et al. (2017), life extension is defined differently, where it is described as extending the lifespan of products and their parts by implementing r-strategies R3-R7 (shown in *Table 1).* However, a common thread in both descriptions is that the life span of the material used is extended through higher resource efficiency and design considerations to promote longer product lives. The definition by Geissdoerfer et al. (2020) is more aimed at extending life spans on a product level, while the description by Potting et al. (2017) is more applicable when analyzing a product on a part/component level. As this thesis will analyze EoL-strategies on a component level, the definition by Potting et al. (2017) will be the one referred to in the thesis.

Life extension is especially important when comparing EVs to internal combustion engine vehicles (ICEVs), where EVs have their biggest environmental impact during the manufacturing phase, while ICEVs emit the most pollutants during the use phase of the life cycle (US Environmental Protection Agency, 2023). As EVs emit less greenhouse gasses during their use phase compared to ICEVs, extending it will improve the total environmental footprint of EVs. It is also important to find ways of extending the product lifespan without compromising the economic viability of the product, hence the importance of knowing when to apply which life extension strategy (Bakker et al., 2014). As different products have

different product characteristics in terms of lifespan, resource intensity and technological maturity, the product life extension strategies will have to be customized to the specific product in question (ibid.). Products and markets change dynamically, which also means that the handling of product life extension will be a dynamic process following the market dynamics, legislation and policy changes (ibid.).

2.4.3 Intensifying

Traditionally, innovation related to sustainability can be highly associated with technical improvements, neglecting people-oriented innovation that treats sustainability as a socio-technical challenge and is affecting elements such as user practices, market meaning, supply network and infrastructure (Adams et al., 2016). In addition to technical innovations, Adams et al. (2016) argue how sustainability-oriented innovation includes adopting new business models or servitization, which affect how innovations are used, how they impact behavior changes, and who they involve. Therefore, in addition to *cycling* and *extension*, *intensifying* is included in the concept of circular business models, which describes how the use phase can be intensified through sharing economy solutions (Geissdorfer, 2020).

Service-oriented business models, often referred to as servitization, can be described as the innovation of organizational capabilities required to shift from selling products to selling product-service systems. PSS can further be explained as a system of services, products, networks of players, and supporting infrastructure that strives to be competitive, satisfy customer needs and have a lower environmental impact than traditional business models (Khan et al., 2020). Products-as-a-service business models are when an organization offers a service, rather than a product, as their value proposition (Lacy & Rutqvist, 2015). Since the service provider owns the products, the organization is responsible for maintenance and repairments during the life cycle instead of the consumers, ideally enhancing reverse logistic services and product life extension. Hence, servitization can be described as a business model for a circular economy (Han et al., 2020).

When implementing servitization in the transportation industry, mobility-as-a-service can be defined as a transition from a paradigm under which mobility is accessed through purchasing a product, to a paradigm where mobility is an outcome of a service (Cooper et al., 2019). Historically, most eco-innovation in the transportation industry have been realized through product modifications and re-design (Machiba, 2011). However, servitization has also been integrated in the transportation industry and contributes the most to sustainable cities by removing the need of vehicle ownership while satisfying the need of users. The need for vehicle ownership can be replaced by bike-sharing services or public transportation (Pinto et al., 2019). An example of an existing service-oriented business model in the transportation industry is the ‘Blue Members’ model implemented by Hyundai (Han et al., 2020). By adopting internet of things (IoT) technology that helps collect data from users, Hyundai has created this model to extend material use and to maximize product life extension by offering their value proposition as a PSS.

2.5 Electric Vehicle Batteries

As the global production of EVs increases, there is an underlying need to establish a well-defined end-of-life procedure specifically for EVs (Ramoni & Zhang, 2013). EVs have a lower environmental impact than ICEVs when comparing total emissions during the life cycle (Xia & Li, 2022). However, EVs have a greater environmental impact than ICEVs during the production phase, mainly due to the production of batteries (Xia & Li, 2022; Del Pero et al., 2018). From an EoL management perspective, the main difference between electric and non-electric vehicles when it comes to recovery of parts are the lithium-ion batteries (LIBs) included in the EVs. An efficient and well-functioning circular system of electric vehicle batteries (EVBs) is therefore a prerequisite for sustainable electro mobility (Glöser-Chahoud et al., 2021). Even though the production of EVBs have been improved in recent years through automation to increase capacity and decrease production cost, EoL treatment of batteries is still simple and underdeveloped through rough manual disassembly (Mossali et al., 2020; Glöser-Chahoud et al., 2021). In an ideal circular economy model, waste LIB-management is integrated with product design, creating systems that favors reuse and recycling and limiting the amount of material to be incinerated or landfilled (Mossali et al., 2020). As recovery strategies for batteries are improving, repurposing and remanufacturing will extend the life of LIBs and recycling will be a significant part from both economic and environmental points of views (Chen et al., 2019). However, as retired LIBs no longer provide satisfactory performance to power EVs, stakeholders can consider less-demanding applications as a second life option (Zhu et al., 2021). Lithium-ion batteries are considered as “green batteries” since they are made of non-toxic materials (Wan & Wang, 2022). However, discarded batteries can be a threat to the environment and human health (ibid.). If lithium-ion batteries overheat and explode when penetrated or when exposed to high temperatures, carbon monoxide and hydrogen cyanide will be released, which can be harmful to human health and the environment. Therefore, it is necessary to recycle lithium-ion batteries efficiently (ibid.).

2.6 Ecodesign & Circular Product Design

The most important decisions for a new product with respect to the environment are made during product development (Go et al., 2015). Ecodesign is a general term that can be defined as product development that enables and facilitates sustainable development. To succeed in eco-designing, integrating environmental competence amongst designers is critical. As requirements for a forthcoming product must coexist with each other, designers must possess environmental knowledge for sustainability to be considered on similar levels as other core engineering principles (Luttropp & Brohammer, 2014). Environmental knowledge and ecodesign is also of great importance for a company’s internal interest as it can be used as guidance to develop more efficient product solutions. Further, ecodesign aims to combine environmental considerations with business-oriented goals to decrease the environmental burden while being financially viable (Karlsson & Luttropp, 2006). The current Ecodesign Directive 2009/125/EC created by the European Union covers 31 product groups and has been proven beneficial for business and the environment both by saving EUR 120 billion in

2021 alone, while decreasing the annual energy consumption for the products in scope by 10 % (European Commission, n.d.-c). Luttropp & Brohammer (2014) created an ecodesign tool called The Ten Golden Principles, which aims to be a guideline for companies to design products with the most environmentally critical factors in mind. The tool gives generic advice for merging environmental aspects into product development.

Circular product design is a methodology focusing on tools and methods that enable the design of products that can be used more than once (Bakker, n.d.). It is based on the Inertia Principle (Stahel, 2010), which is a notion that a product should, in principle, never become waste. The goal is always to preserve a product in its initial state over time, which is ultimately an utopian vision in which the product integrity is never compromised (Hollander et al., 2017). However, in reality it is understood that a product will move down the waste hierarchy sooner or later, but circular product design follows the principle of a product staying in its original state over multiple lifecycles. Thereby, recycling is an undesirable option from a product integrity perspective given that it involves the dismantling and disintegration of a product and its constituent components and the subsequent reprocessing of the product's materials (Hollander et al., 2017).

The key difference between the methodologies of ecodesign and circular product design is its different views on the product life cycle. Ecodesign aims to improve the environmental performance of a product life cycle, and from an ecodesign perspective the life cycle follows the waste hierarchy model where the current definitions of prevention, reuse, recovery, and recycling all assume that a product ultimately ends up as waste (Hollander et al., 2017). Circular product design is rather aimed at designing a product that can be used in multiple lifecycles by reversing obsolescence at a product, component, and material level (Asif et al., 2021).

2.6.1 Design for X

Design for X (DfX) is an umbrella term for design strategies created to capture the knowledge of experts and give designers further possibilities to improve designs (Jack, 2021). There is no single body for "design for"-techniques, but several are listed in *Table 3* below. Although, as effort towards reducing total life cycle cost through design innovation has become an essential part of the manufacturing industry, DfX are also associated with design for environment and design for recyclability (Kuo et al., 2001). Also, concurrent engineering and DfX are associated with product development, revealing great potential to enhance different service functionalities like circularity (Sassanelli et al., 2021). Since EoL strategies may include a combination of R-strategies, designers must identify ideal EoL strategies prior to designing a product to avoid costly product retirements. When integrating DfX techniques into product design for sustainability, several tools have been developed and improved to improve the environmental impact of products. *Design for Disassembly (DfD), Design for Modularity (DfM),* and *Design for Recycling (DfR)* are some green design tools used to lower emissions in all life cycle stages of products (Go et al., 2015).

Table 2: DfX-types.

DfX Acronym	Meaning
DfD	Design for Disassembly
DfR	Design for Recycling
DfS	Design for Sustainability
DfQ	Design for Quality
DfR	Design for Reuse
DfM	Design for Maintenance

2.6.2 Design for Disassembly

Disassemblability is a prerequisite for product components to guarantee recovery of material and parts, where recovery of ELVs is environmentally and financially beneficial for the dismantlers (Favi et al., 2019; Petrauskiené et al., 2022). According to Go et al. (2011), reuse, remanufacturing, and recycling are three EoL-options that can be categorized as recovery strategies. When considering the different strategies, core quality of used products, market value, and implementation cost for the different strategies must be taken into consideration (Östlin et al., 2009). However, as driving forces and barriers based on technological and economical motives vary between components, a single recovery strategy for an assembly is difficult to implement (Östlin et al., 2009). Hence, the environmental impact from automotive components can only be decreased if the products can be disassembled, where the cost effectiveness of recycling increases as the disassembly of automotive parts is simplified (Afrinaldi et al., 2008; Favi et al., 2019). Further, to achieve the goal of closing material loops and creating economic activities based on loops, vehicles need to be designed as technical systems based on modular master plans with ease of disassembly and ease of maintenance (Rios et al., 2015; Bellmann & Khare, 2020).

Design for disassembly is a method that can be applied during the design stages to improve overall product recovery (Go et al., 2011). According to Bellmann & Khare (2000), designers of vehicles give little to no thought on design for end-of-life management since most companies lose responsibility for vehicles once they are sold. However, integrating end-of-life strategies in the early stages of design is the most important aspect to improve recovery of ELVs and automotive components to determine environmental and economic benefits (Go et al., 2011; Petrauskiené et al., 2022). DfD is a design methodology to ease

deconstruction and support engineers in developing products that are easily dismounted into single components to reduce disassembly time/cost and recover relevant quantities from components (Rios et al., 2015; Favi et al., 2019). DfD involves component design, material selection and product architecture to support product recovery (Bovea et al., 2016). When modules of ELVs have been disassembled to single components, a suitable end-of-life strategy can be implemented for each component to create a more sustainable vehicle (Go et al., 2011; Östlin et al., 2009). Further, products designed using the DfD method can deliver greater savings than products designed without this intention as repairments, and services become simplified and more efficient. Consequently, implementing DfD is an additional economical driving force for organizations in the transportation industry (Go et al., 2011).

The transition towards sustainable practices with lightweight electric vehicles has led to an increased use of aluminum in the transportation industry (Soo et al., 2018). Despite being one of the most recycled metals, ongoing multi-material designs and unliberated joints increases the difficulty of separating aluminum with high purity (Soo et al., 2018; Soo et al., 2019). Further, the complexity of recycling aluminum compared to other metals comes from the relatively low melting point of aluminum, which results in difficulties in removing impurities or tramp elements (Soo et al., 2018). Most of the design rules specific to joining choices correspond to the theories of DFD when reuse and remanufacturing are the designated recovery strategy (Go et al,. 2011; Soo et al., 2018). However, the type of joint technique depends on the suitable recovery strategy for each component. While mechanical joints such as steel rivets and bolt screws are favorable when considering remanufacturing and reusing components due simplifying disassembly, the same joint techniques cause high impurities when recycling aluminum (Soo et al., 2019). When recycling aluminum, adhesive bonding decreases impurities of aluminum compared to mechanical joints (Soo et al, 2018). To increase purity of aluminum when recovering components of vehicles, joint techniques should be customized for different life cycle scenarios (Soo et al., 2018).

Among all the documented studies regarding Design for X, it can be observed that DfD is the least commonly addressed (Go et al., 2011). However, manufacturers of vehicles must consider DfD in their new design as increasing the level of remanufacturing and reuse is a key part of moving towards a sustainable production of vehicles. Within the academic community, there are few efforts to rationalize design for disassembly, where most research concerning these issues have been performed in the 1990s (Go et al., 2011). Designed by McGlothlin and Kroll (1995) and improved by Kroll and Hanft (1998), the spread sheet-like disassembly evaluation chart was created to measure ease of disassembly. Each task is evaluated using difficulty scores for each difficulty based on a scale of 1 (easy) to 10 (difficult) on various conditions (Fatmawati & Razak, 2007). The various conditions are described in the first six columns of the spreadsheet as a reference to the difficulty ratings, where the difficulties were categorized between column 7-11, as *accessibility*, *positioning*, *force*, *base time* and *special* (See *Figure 3* below).

- *Accessibility*: a measure of the ease with which a part can be reached by hand or by the tool during disassembly and how easily a part can be maneuvered during this stage.
- *Force*: a measure of required force during disassembly.
- *Positioning:* a measure of the required precision of tools to disassemble a product or module.
- *Base time:* refers to the time required to do the basic task movements without difficulty.
- *Special:* A measure included to note special problems that is not included in any other category.

DISASSEMBLY EVALUATION CHART

PRODUCT: DATE:

PREPARED BY: SHEET:

1	2	3	4	5	6	7	8	9	10	11	12	13	14
Part Number	Quality	Theoretical Minimum Number of Parts	Task Type	Number of Task Repetition	Required Tool	Difficulty Rating 1(EASY)...10(DIFFICULT)					SUBTOTAL	TOTAL	COMMENTS
						accessibility	positioning	force	additional Time	special			

Figure 3. Disassembly Evaluation Chart (Go et al., 2011)

When completed, weaknesses in design can be recognised based on the ratings from the evaluation chart. Two metrics which can be calculated are the design effectiveness of each product and an estimation of the disassembly time. The design effectiveness is defined as follows:

$$Design\ effectiveness\ = \frac{5\ x\ \Sigma\ Theoretical\ Minimum\ Number\ of\ Parts}{\Sigma\ Total}\ x\ 100 \quad (1)$$

Equation (1) compares the disassembly difficulty of the current design to an ideal design of the same product where 5 seconds is the ideal disassembly time per part (Fatmawati and Razak, 2007; Kroll and Hanft, 1998). Meanwhile, the disassembly time, converted to seconds where *Number of Tool and Hand Manipulations* is accumulated from column 5 and column 6 (See *Figure 3*), can be calculated according to the following equation:

$$Disassembly\ time\ = (\Sigma\ Total\ -\ 5\ x\ \Sigma\ Number\ of\ Task\ Repetition)\ x\ 1.04\ +\ldots$$
$$\ldots\ (Number\ of\ Tool\ and\ Hand\ Manipulations)\ x\ 0.9 \quad (2)$$

The main goal of the spread sheet-like chart is to decrease the complexity of disassembly (Fatmawati and Razak, 2007; Kroll and Hanft, 1998; Go et al., 2011). The results from the evaluation, such as design efficiency, and disassembly time estimation are calculated and then

evaluated to identify areas that could be improved. Optimizing design is achieved by analyzing the evaluation results, making improvements on the current design and finally re-evaluate the design (Go et al., 2011). The spread sheet-like chart can be implemented as a tool to rationalize DfD (Fatmawati and Razak, 2007; Kroll and Hanft, 1998; Go et al., 2011).

2.6.3 Design for Recycling

Design for Recycling is a design method consisting of several tools created to optimize recycling (Veelaert et al., 2017). Regarding the automobile industry, the current ELV processing can be divided in two sections: dismantling and shredding. DfR aims to simplify disassembly of product parts as it can increase the recycling rates of products and limit shredding of vehicles. However, while dismantling is a key activity in DfR, to limit the labor-intensive dismantling processes of vehicles, organizations within the transportation industry have invested in technologies to improve recycling efficiency of vehicle shredder residue (Ferrão & Amaral, 2007). Nevertheless, focusing on optimizing recycling efficiency from shredded vehicles only neglects more sustainable options in the waste management hierarchy model (See *Figure 2*). According to Afrinaldi et al. (2008) and Favi et al. (2019), disassembly is crucial to limit the environmental impact from vehicle components since a variety of EoL-applications are enhanced by the dismantling of the product parts. While DfD promotes a simplified disassembly to enhance several EoL strategies, including the simplification of separating materials for recycling purposes (Go et al., 2011), DfR aims to improve recycling processes only. Hence, adapting the methods and tools from DfR while excluding other DfX types can increase recycling rates but also neglect more sustainable EoL-applications. However, despite prevention and reuse being EoL-applications that result in more environmental benefits than recycling (Korhonen et al., 2018), recycling is the only feasible option for some vehicle components. For components where recycling is the optimal EoL-application, the recycling rate, the cost of recycling and the characteristics of the local recycling infrastructure must be taken in consideration during the design stages of vehicles (Ferrão & Amaral, 2007). As the aforementioned aspects are covered in the concept of DfR, implementing this design strategy can help improve EoL-management.

2.6.4 Design for Modularization

A modular design can be defined as a design that allows a number of modules through defined interfaces to compose a variety of products, and modularity aims to break the architecture of a product into independent units (Sonego et al., 2018). Further, product modules can be defined as integral product substructures that have an individual correspondence to a product's functional model. Implementing Design for Modularity and making products modular may enhance remanufacturing capabilities. Due to the functional independence in a modular product's configuration, modularity can make reuse, remanufacturing and maintenance simpler (Go et al., 2015). Additionally, modularization gives possibilities to reduce the total number of parts in a variety of similar products due to them having shared product substructures (Luttropp, 2000). As a result, modular designs increase the chances of finding second-life applications for product components as the same modules can be applied in a variety of products. While modularization is commonly

addressed by academic research to lower environmental impact in the production phase, Sonego et al. (2018) argue how modularization can claim benefits in all phases of a products life cycle, a concept called green modularization. Green modularization is a way to perform modularization from a life cycle point of view, and not only with functional and structural requirements (ibid.). To clarify, while modularization is typically applied to favor functionality and to simplify production, it can also be applied with life cycle and post life intent. For instance, as modularization favors remanufacturing, replacement and upgrades of components, modularization copes with the concept of multiple product life cycles while limiting the risks of product obsolescence over time, conforming to *Figure 4* below (Asif et al., 2021).

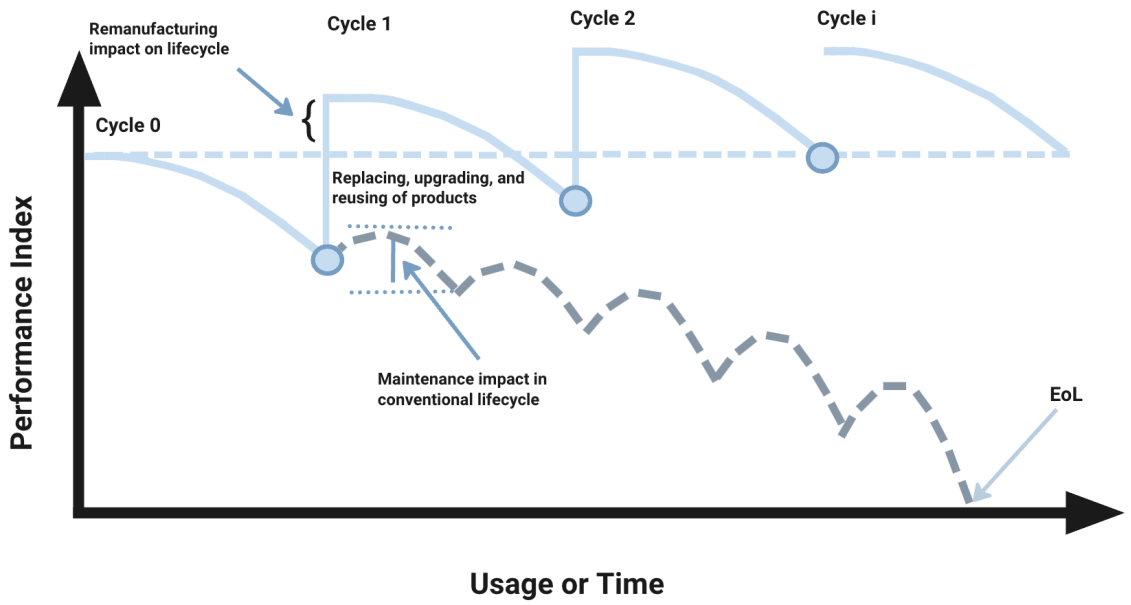


Figure 4: Replacing, upgrading and reusing to promote multiple product life cycles, based on Asif et al. (2021).

Several methods of green modularization methods and methods that concern life cycles have been established. Newcomb et al. (1996) presented one early method which concerns functionality, physical appearance, and EoL. Since then, additional methods have been established to cover additional aspects related to Ecodesign. Among others, Gu and Sosale (1999) presented a modularization method for life cycle engineering that concerns services, assembly, reuse, and recycling at the design stage. An additional method was presented by Smith and Yen in 2010, where products are organized by sustainability constraints, such as disassemblability, material compatibility and recyclability (Smith & Yen, 2010).

3. Method

In this chapter, the innovation processes used during the thesis work are presented. This chapter also presents a description of how primary and secondary data regarding EoL-management was accumulated.

3.1 Double Diamond Design Process

The Double Diamond design process has been applied in this project as it aims to achieve innovative solutions, focusing on designing the right things and designing things right. This innovation process consists of four stages, being *Discover, Define, Develop,* and *Deliver,* see *Figure 5* below. The first diamond of the process, defined as the problem space, focuses on discovering and defining the problems related to the project in order to design an appropriate solution. The second diamond, defined as the solution space, focuses on developing and delivering a solution to the problem in the best possible way (Gustafsson, 2019). Further, Iterative prototyping has been used in the second diamond of the design process, as it aims to quickly make ideas tangible, making ideas easier to evaluate and refine (Brown & Katz, 2009). Since the solution in this case is a set of processes and guidelines rather than a product, prototyping in this project relates to finding conceptual solutions to different parts of the project scope.

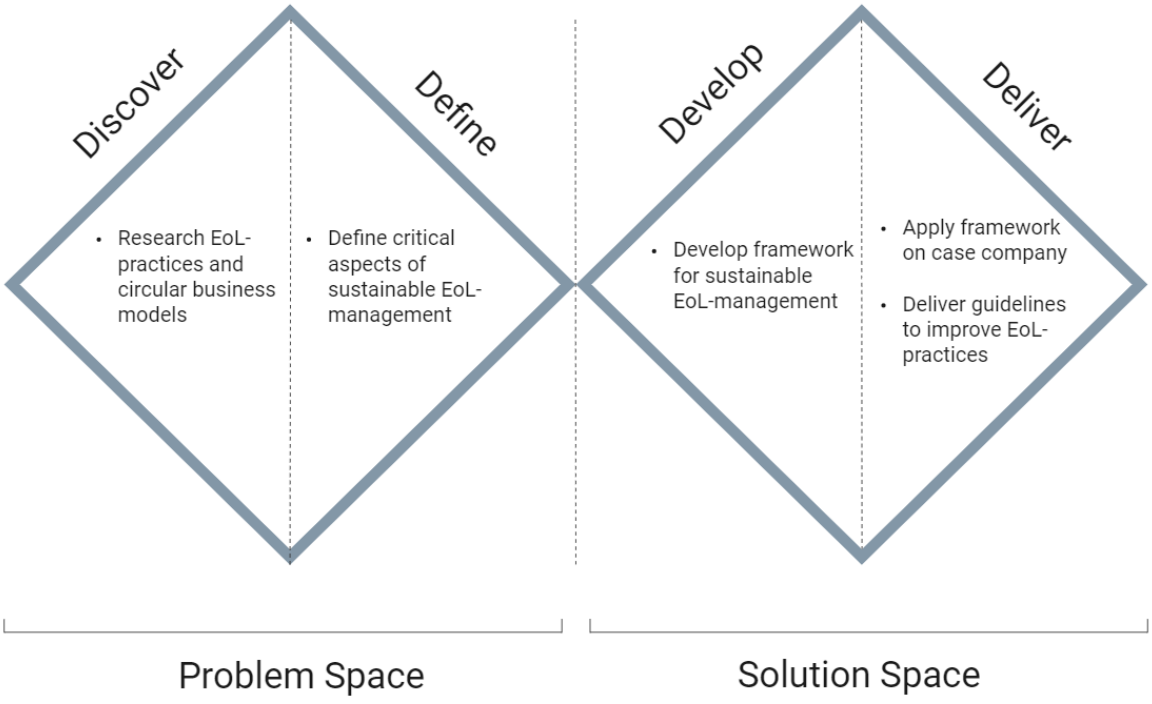


Figure 5: Double Diamond Design Process, based on Gustafsson (2019).

3.2 Secondary Data Collection

To get a broader understanding of the research area, secondary data was acquired to assess and evaluate previous work conducted within the field. This included examining existing

literature regarding ELV-management as well as investigating the prerequisites for a functioning EoL-management system.

3.2.1 Literature Review

The literature review was conducted by establishing key research areas within the field of ELV-management. Different databases such as Google Scholar, KTH Primo, Scopus and Web of Science were browsed with key words from the identified areas of importance. The key words used were *End-of-Life Vehicles, ELV-Management, Reverse Logistics, End-of-Life Motorcycles, Recovery of Vehicles, R-strategies, Li-ion Battery Recycling, Design for X, Circular Business Models*. The selected topics were verified with supervisors from both the case company and KTH to see if they were relevant to the thesis scope.

3.3 Primary Data Collection

Primary data was collected to fill knowledge gaps from prior studies regarding ELV management and receive both qualitative and quantitative data to provide a comprehensive understanding of the subject matter. Additionally, primary data was collected to receive further understanding of how CAKE's products are managed during the life cycle, and how logistics are managed for all components. Internal and external interviews, visits and other observations were obtained to collect primary data.

3.3.1 Interviews

To receive further understanding of end-of-life strategies used in electric two-wheelers, interviews were held with experts of different areas. The purpose of the interviews was to understand critical aspects of CE and key activities regarding EoL management in the transportation industry. Additionally, a secondary purpose of the interviews was to receive the experts' opinions regarding the conducted literature review. All interviews were based on semi-structured interview guides concerning *R-strategies, CE, Current practises* and *Design for X*. Semi-structured interviews have been applied since the method is epistemologically versatile and compatible with qualitative and quantitative data (McIntosh & Morse, 2015). Further, semi-structured interviews are beneficial as it gives the respondent the opportunity to express themselves freely and enable the opportunity for the respondents to give additional inputs that can benefit the research (Denscombe, 2017).

Table 3 shows a summary of all interviews, describing the profession of each respondent, type of interview, the time duration and the type of documentation. The type of interview altered between digital interviews via Google Meet, Zoom or Microsoft Teams, and physical meetings with the respondents. The time duration of the semi structured interviews were between 30-90 minutes. The interviews were documented by recording the audio or by taking notes during the meetings. The reason why not all the meetings were recorded was because it was seen as more appropriate to take notes in physical meetings, while recording was deemed as more appropriate in digital meetings. According to (Guest et al., 2017), individual interviews can be highly effective at generating items in a brainstorming task. Hence, all interviews were conducted individually with the respondents. Lastly, all interviews were

conducted with open ended questions to decrease the risk of different biases (Reja et al., 2003).

Table 3: Interviews and meetings conducted during the thesis work.

Respondent	Role, Location	Type of interview	Time (min)	Documentation	Date
Respondent 1 (Internal)	Internal Service Team, Stockholm	Physical meeting at CAKE	90	Notes	2023-02-22 2023-03-14 2023-04-04
Respondent 2 (Internal)	Internal Product Team, Stockholm	Physical meeting at CAKE	45	Notes	2023-02-28
Respondent 3 (External)	R&D Portfolio Manager, Stockholm	Microsoft Teams	60	Notes	2023-02-28
Respondent 4 (External)	Sustainability, Energy and Environmental Manager, Stockholm	Zoom	45	Notes	2023-03-01
Respondent 5 (External)	Professor in CE, Stockholm	Zoom	45	Audio recording	2023-03-07
Respondent 6 (External)	Sustainability Manager, Uppsala	Zoom	30	Audio recording	2023-03-08
Respondent 7 (External)	Professor in Ecodesign, Stockholm	Zoom	65	Audio recording	2023-03-10
Respondent 8 (External)	Senior Researcher in Sustainable Transport Design, Stockholm	Zoom	45	Notes	2023-03-16
Respondent 9 (External)	Vice President of Modularization Firm, Stockholm	Physical Meeting	60	Notes	2023-03-17
Respondent 10 (External)	Sales and Product Manager, Lennartsfors	Zoom	30	Notes	2023-04-04
Respondent 11 (External)	Senior Sustainability Engineer, Göteborg	Google Meets	30	Notes	2023-04-11
Respondent 12 (Internal)	Customer Offers Manager, Stockholm	Google Meets	30	Notes	2023-04-19

3.3.2 Visits & Observations

To get a more practical approach to how an End-of-Life management system could be established, physical visits to relevant locations have been made. During these visits, observations were made to get a broader understanding of the problem scope. The visited locations can be seen in *Table 4* below.

Table 4: Visits organized during the thesis work.

Location	Time (min)	Documentation	Date
CAKE Assembly Site, Albyberg	Full day	Field notes & pictures	2023-02-17
CAKEsite :Stockholm, Stockholm	60 x 2	Field notes	2023-02-22, 2023-04-04

CAKEsite :Stockholm is CAKE's own Showroom/Retail/Test Ride & Service Space, which was visited to speak to the service team about which parts of the motorbikes are replaced most often, how they could be better designed for End-of-Life management, what repairability errands the service team can and cannot handle, which parts could be reused/refurbished and more. The visit to the CAKE Assembly Site was made to weigh the different components of the Makka:work bike to see how much of each material it consisted of. Another reason was to inspect how the bikes were assembled to investigate if design changes could be made to facilitate repairability and disassembly. During the visits, field notes and pictures were taken to have documentation of important observations which could be needed in a later stage of the thesis work.

4. Development of Framework

This section covers how a framework for a sustainable EoL-management system has been developed. Its contents are both based on primary and secondary data, and different evaluation methods have been chosen in respect to what are seen as critical factors when establishing an EoL-management system. An overview of the framework is illustrated in *Figure 6*. The included aspects of the framework are described in detail in the following sections.

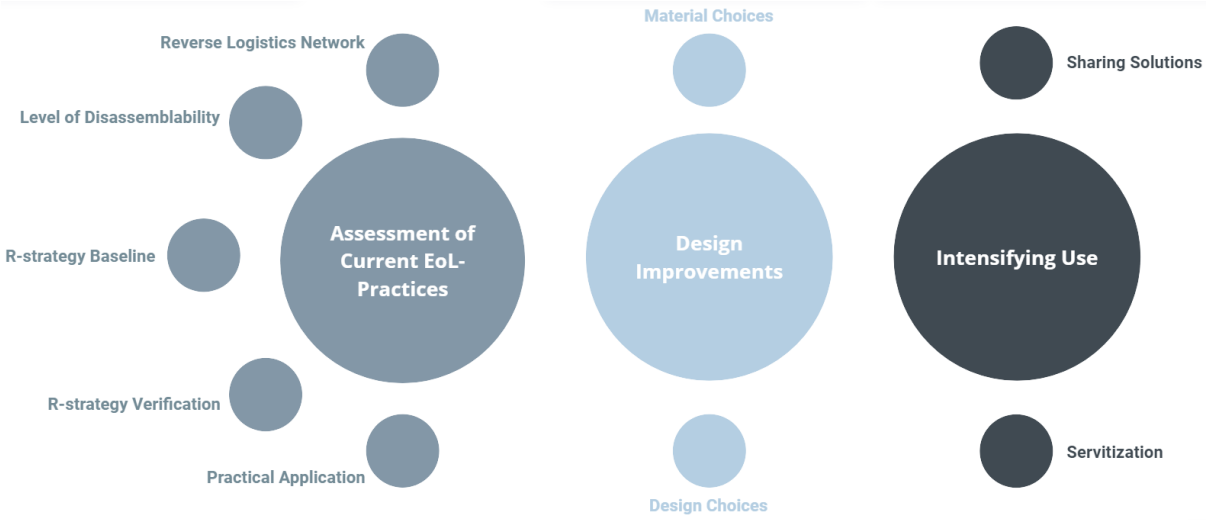


Figure 6: Three-stage framework for End-of-Life management.

4.1 Assessment of Current EoL-strategies

To get a sense of where the company is at with its existing practices, the current EoL-strategies should be evaluated in at least five different aspects. Based on the findings from primary data and secondary data, the most critical factors in this framework are seen to be the following: *Reverse Logistics Network, Level of Disassembly, R-strategy Baseline, R-strategy Verification and Practical Application,* see *Figure 7* below.



Figure 7: Factors included in the assessment of current EoL-practices.

These aspects can be seen as the baseline of estimating how EoL-strategies can be implemented currently, and once these aspects have been analyzed, more detailed assessments in strategic and operational elements can be implemented.

4.1.1 Reverse Logistics Network

The reverse logistics (RL) network is essential for end-of-life management. First, it should be established if the company has a way of returning its products from the consumers in the EoL-phase, whether it is by implementation of collection systems, getback or buyback systems or by using business models maintaining producer responsibility of the product through its entire lifecycle. Secondly, the firm's warehousing capabilities should be analyzed in terms of inventory management, sorting and inspection (Wilson & Goffnett, 2022). In addition, strategic tools and evaluation methods should be applied, such as cost benefit analysis (CBA), secondary market forecasting and investigation of environmental regulations (ibid.). This should be done to get a more comprehensive view of how different important aspects of RL-network would function in practice. Further, the managerial aspects of the reverse logistics network should be analyzed with strategic, tactical and operational considerations in line with *Figure 8* below.



Figure 8: Managerial considerations of RL-network, based on (Wilson & Goffnett, 2022).

4.1.2 Level of Disassembly

A weighted point system was created to assess the level of disassembly for all ingoing parts of a product. Based on primary data from Respondent 7 and secondary data presented by Go et al. (2011), the weighted point system was created to cover the information needed, tools necessary, and accessibility necessary to dismantle a part from an electric two-wheeler. Further, all components received a score (0, 0.5, or 1) in these categories, summarizing the overall disassembly level of a component. The scores were determined to be a numeric interpretation of the difficulty level in each category. A low overall score was concluded as an easy disassembly while a high score was concluded as more difficult. A combined score below 1.5 was decided to give the product part the classification of *Easy*, and a score below 2 would classify the part as *Medium*. Lastly, a combined score above or equal to 2 would classify the part as *Difficult*. The descriptions of all categories are explained in *Table 5* below.

Table 5: Weighted point system for disassembly.

	0	0.5	1
Information	No information needs to be provided. Everyone understands how to disassemble part	Requires assembly instructions	Help from experienced service members is needed
Tools	No tools are needed	Simple tools are needed (eg. screwdriver, hex key etc.)	Special tools are needed
Accessibility	Easy to reach	Medium accessibility	Hard to reach (may require removing other components to disassemble)

4.1.3 R-strategy Baseline

To select which r-strategy is suitable for each component, several factors must be considered. In *Table 6* below, all r-strategies are presented along with their waste hierarchy categorizations and the questions posed as they are analyzed. However, conforming to Bakker et al. (2014), the posed questions were customized to fit an analysis on a component level. To analyze the potential circular flow of material used in a product, the hierarchical categorization *Reuse (r-strategies R3-R7)* shown in *Table 6* are used to categorize how many parts have a second life potential (Bakker et al., 2014).

Table 6: Description of how an r-strategy is chosen for each part of a product.

Hierarchical categorization	R-strategy	Analysis question
Prevention	R0 Refuse	Is the part necessary for functionality?
	R1 Rethink	Is the part manufactured and used in an ideal way?
	R2 Reduce	Does the part use an unnecessary amount of natural resources?
Reuse	R3 Reuse	Is the part still in good condition to fulfill its original function?
	R4 Repair	Can the part be repaired by service team employees, and is it economically justifiable?
	R5 Refurbish	Can the part be restored to fulfill its original function?
	R6 Remanufacture	Can the part be restored to the level of being used in a completely new product with the same function?
	R7 Repurpose	Can the part be used in another application with a different function?
Recycle	R8 Recycle	Is the material recyclable?
Recovery	R9 Recovery	Is there energy to be recovered or should part go to landfill?

4.1.4 R-strategy Verification

An important aspect when assessing whether the correct r-strategy in the hierarchy was chosen is to question how the conclusion came about. It is easy to have a preconceived idea of which r-strategies should be applied for specific parts, but when questioned, there could be a realization that the parts can climb even higher up in the hierarchy. This notion is partly based on the Inertia Principle by Stahel (2010), where he wrote: *"Do not repair what is not*

broken, do not remanufacture something that can be repaired, do not recycle a product that can be remanufactured. Replace or treat only the smallest possible part in order to maintain the existing economic value". In line with this thought process, after an r-strategy is chosen for each part, the question of *"why no higher r-strategy is chosen?"* should be posed. In many cases, most parts could technically be recycled for example, but choosing recycling as the r-strategy on all those parts would perhaps disregard r-strategies higher up the hierarchy (Potting et al., 2017). The same would go for repairing parts that could be rethought, reusing parts that could be refused etc.

4.1.5 Practical Application

The final part of assessing the current EoL-practices is to analyze how the r-strategies could practically be applied by the company. As every company and their products have different conditions and constraints, this answer will likely differ a lot depending on who is carrying out the assessment. Therefore, it is important to consider the type of product, the company resources, and the business model of the company making the assessment. The practical application should also be in line with the company vision regarding sustainability and circular economy.

4.2 Design Improvements

To move towards more sustainable EoL practices in line with the waste management hierarchy model (see *Figure 1*), analyzing the product design to improve its environmental impact is seen as an important factor of the framework. These improvements aim to enhance circulation of material and promote product life extension, corresponding to the concept of CE. The improvements in design can further be categorized as *material choices* and *design choices*, which can be seen in *Figure 9* below.

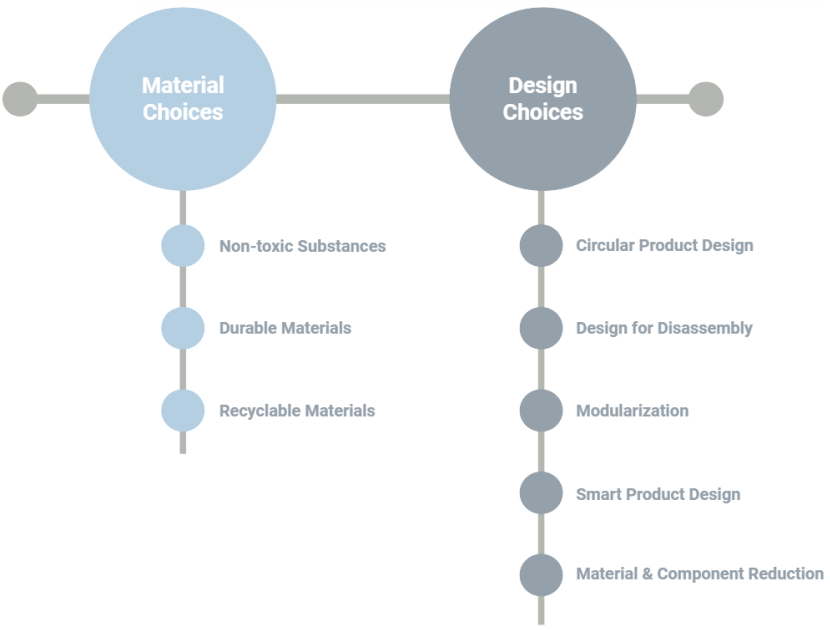


Figure 9: Factors included in analyzing product design improvements.

4.2.1 Material Choices

As the material choices can affect EoL-options and the life expectancy of a product, poor choices of materials when designing a product can neglect more sustainable practices during all stages of a product lifetime (Go et al., 2011; Bakker et al., 2014). To improve the material choice from a sustainability perspective, designers and manufacturers should consider the following three aspects: *Non-toxic substances, durable materials,* and *recyclable materials*.

Using ***non-toxic substances*** (substances that are not dangerous and cannot cause injuries (Yuan & Dornfield, 2009)) is essential in ecodesign to create circular flows of materials and to find EoL-options high up in the waste management hierarchy model. In accordance with the ten golden principles for ecodesign (Luttropp & Brohammer, 2014), developers and manufacturers of vehicles should consider finding substitutes for non-toxic substances which fulfill similar requirements as the previous alternatives (Luttropp & Brohammer, 2014; Go et al., 2011). However, if completely removing hazardous substances is unfeasible, keeping these substances in a closed loop is essential (Luttropp & Lagerstedt, 2006).

Durable materials refer to materials that can cope well with the load cases the product is intended to be exposed to, minimizing the risk of damaged material and the need of service (Cooper, 1994). To prolong the product life expectancy and enhance more sustainable EoL-options, more durable materials should be considered, especially in parts that need a lot of service during the product's life cycle. As more durable materials extend the life expectancy of product parts, durable materials can help move towards more sustainable product life cycles by circumventing physical product obsolescence (Rossi et al., 2022). Durability does not only have functional benefits as durable materials are also less prone to receiving aesthetic damages such as scratches and visible deteriorations (Bakker et al., 2014).

Recyclable materials are materials that have been diverted from non-hazardous waste streams for the purpose of reuse, recycling or reclamation (City of Decatur, IL, 2015). Using recyclable materials means avoiding materials where the EoL-options are limited to energy recovery or disposal. As previously mentioned, recycling is seen as undesirable from a circular product design perspective as extended product life and multiple product life cycles are preferable according to the philosophy. However, as all materials eventually become obsolete due to wear, tear, stress or other factors affecting the material properties, using materials that can be recycled at the end of their usable life is seen as advantageous from an environmental perspective.

According to a professor in CE (Interview Respondent 5; March 7, 2023), a tradeoff can be had between having durable materials that are non-recyclable or to use less durable materials that are recyclable. In many cases, using non-recyclable materials that instead promote longer product life can be preferred from an environmental perspective according to the respondent.

4.2.2 Design Choices

According to Go et al. (2011), design choices have a direct impact on EoL-management. Hence, making changes in design can lower the environmental impact from vehicles and enhance CE practices. Based on findings from the respondents and from secondary data, information regarding changes in design to improve EoL-management have been divided into five categories: *Circular product design, Design for disassembly, Modularization, Smart product design,* and *Material & component reduction*.

Circular product design refers to adapting the design of a product to last for multiple lifecycles (Bakker, n.d.). To improve the rate of second life potential, manufacturers of vehicles should strive to preserve the product's initial state over time, which in turn simplifies EoL-management. Since preserving the initial value of vehicles limits the need of maintenance between or during life cycles, a circular product design enhances preferred EoL-options in the waste management hierarchy model (shown in *Figure 1*).

A vehicle should be ***designed for disassembly*** to facilitate the EoL-management of the product. To implement EoL-options such as refurbishing, reusing or repairing, subassemblies of a vehicle need to be disassembled and managed as an individual system, since driving forces and barriers vary for different subsystems (Go et al., 2011). However, for disassembly of vehicles to be encouraged, it must be possible to implement in an efficient manner, limiting the cost and the need of an organization's capabilities. By implementing design for disassembly, dismantling of product substructures are simplified which favors product life extension and EoL-management.

The implementation of ***modular systems*** can help standardize the manufacturing and service of ingoing product parts, simplifying the production line while increasing remanufacturing capabilities. Thus, modular systems have a positive impact on EoL-management. Further, finding second life options is simplified when using modular systems between several vehicles, either within one organization or as a collaboration between different manufacturers (Asif et al., 2021). Lastly, as modular systems focus on physical interfaces between different subsystems, a modular design may simplify disassembly, which in turn favors all r-strategies in the hierarchical categorization *Reuse* (shown in *Table 6* in section *4.1.3*).

Smart product design refers to improving the design of product substructures with sustainability and EoL-management in consideration. By developing alternate solutions to the same functional requirements, some solutions may affect the environment less and might simplify EoL-management more than others. Therefore, design changes conforming to this category can be applied on components which receive *R1 Rethink* as a suitable r-strategy. However, smart product design should not be limited to one r-strategy, as alternative solutions that affect the environment less than current solutions can be discovered for most components. By considering alternative designs for similar functions, companies within the transportation industry have an opportunity to extend component lifetimes, decrease the need of maintenance and simplify manufacturing processes by using smarter design principles.

Material and component reduction refers to removing parts of a vehicle that does not add customer values nor add value to the vehicle's functional model. By removing unnecessary materials and components from the vehicle, the complexity of the products and the EoL-management both become simplified as less material and fewer substructures are used. Further, since less material is used in the vehicle, less energy is required through all stages of the product's life cycle. Therefore, considering removing certain parts due to lack of functional contribution should be implemented by developers of vehicles.

4.3 Intensifying

Circular business models are realized through both technical and socio-technical innovations, including changes in value proposition, value creation, and value capture (Geissdoerfer et al., 2020; Adams et al., 2016). To intensify the use phase of electric-vehicles and help change the social behavior of users, manufacturers should consider moving towards shared vehicle systems over personal ownership of the same products. Additionally, as shared vehicle systems give companies ownership of the products, these systems enhance servitization which in turn has a positive impact on product life extension and EoL-management (Han et al., 2020). To fulfill the customer needs while minimizing the production of vehicles through different shared vehicle systems, a question for organizations to ask themselves is "*How can one vehicle be used by as many consumers as possible while gaining similar profits as selling one vehicle to each customer?"*. By adopting the type of revenue model to the capabilities and needs of each company, organizations within the transportation industry have an opportunity to enhance circular business models on different levels and become increasingly more sustainable.

5. Case: Framework Implementation at CAKE

In this chapter, the results obtained when implementing the framework at the case company CAKE is presented. It should be noted that not all aspects of the framework are relevant to CAKE as some might already be implemented by the company.

5.1 Assessment of Current EoL-strategies

The assessment of CAKE's current capabilities of managing EoL for the motorbike Makka:work is presented below.

5.1.1 Reverse Logistics Network

CAKE has a reverse logistics network today which includes business offers such as leasing and subscriptions where they maintain product responsibility through the motorbike's entire lifecycle. For the products that are sold, they have a buyback-system where they can guarantee the consumer a purchase price for the motorbikes based on their driven distance and condition. These bikes are later refurbished and sold on CAKE's own second hand platform called re:CAKE. As CAKE already has an established and functioning reverse logistics network on the Swedish market, no specific improvement proposals for the company are presented within the area.

5.1.2 Level of Disassembly

Based on the three categories *information*, *tools*, and *accessibility* in the disassembly evaluation method, every product component of Makka:work received a total score to constitute their level of disassembly. The evaluation was made from inputs from the service teams as well as a disassembly instruction manual of Makka:work received by CAKE. The scores reflect whether the disassembly levels of the components are seen as easy, medium or difficult to disassemble. An example of three components receiving three different levels of disassembly can be seen in *Table 7* below. Since the frame is the foundation where most parts are attached, dismantling the frame is practically the same as removing all parts except the frame from the electric two-wheeler. As a result, this product part is given the highest level in all categories. The rear light requires basic tools but does not require any further parts to be removed or any instructions of the dismantling process, making it easy to disassemble. The steering spacer also requires simple tools to be removed. However, several parts like the upper stem, lower stem and the handlebar need to be removed in order to dismantle the steering spacer, which results in the steering spacer being classified as a medium level of disassembly.

Table 7: Practical application of evaluation method for level of disassembly.

Name	Index			Tot	Note	Level of Disassembly
	Info	Tools	Access ability			
Rear light	0	0.5	0	0.5	Simple tools are needed	Easy
Steering spacer	0	0.5	1	1.5	Simple tools and removal of other parts are necessary	Medium
Frame Makka	1	1	1	3	All components need to be removed from the frame	Difficult

The evaluation of the level of disassembly concerning components from the bike was performed on all of the product's components. A vast majority of the ingoing parts of Makka:work are easy to disassemble, which shows that the motorbike is already designed with DfD-principles in mind. The distribution between product parts that are easy, medium and difficult to disassemble from the electric two-wheeler can be seen in *Figure 10* below.

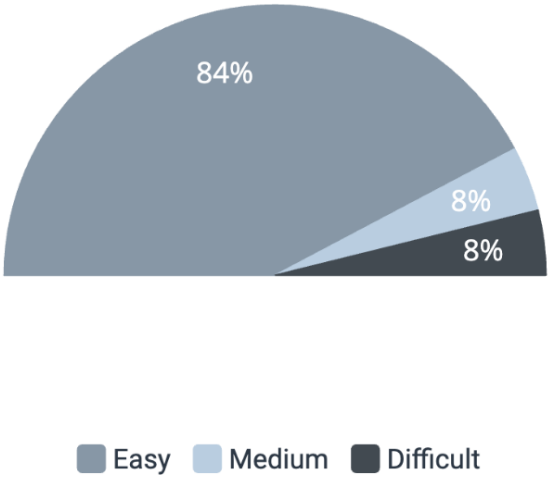


Figure 10: Distribution between different levels of disassembly.

5.1.3 R-strategy Baseline

Based on the r-strategy analysis described in *section 4.1.3*, a suitable r-strategy was chosen for every single part of the motorbike Makka:work. As the r-strategy evaluation was based on the lifespan of 20 years or 600 000 travelled kilometers, the components had to be evaluated in terms of material, quality and functionality. In some cases, the economic and practical

feasibility also had to be considered. As CAKE are not the original manufacturers of many of the components, there is a lack of competence in how to repair some of the parts. Even in cases where repair might be practically possible, the economic justification could be insufficient. The distribution between the selected strategies for all parts can be seen in *Figure 11* below. Note that *R0 Refuse*, *R2 Reduce*, *R4 Repair* and *R6 Remanufacture* are not included in the chart as these were not seen as suitable strategies for any part of the bike. For verification purposes, members from the service team were also involved in assessing which parts need to be replaced most often, as well as verifying the chosen r-strategies once the analysis was complete.

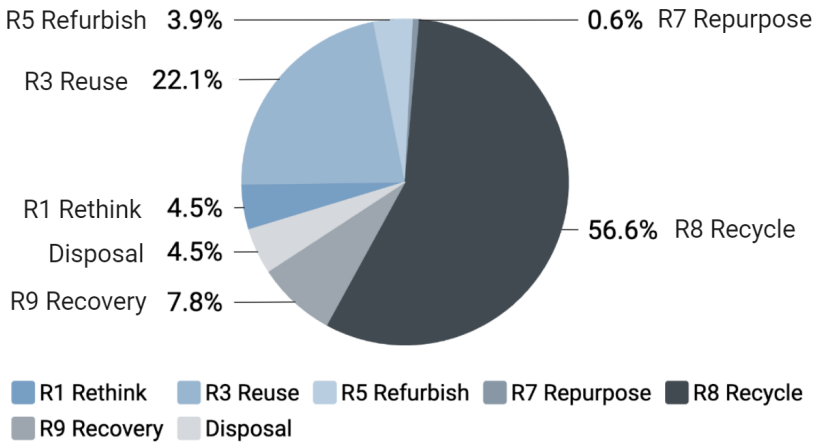


Figure 11: R-strategy Baseline, part distribution.

To get a better estimate of how the r-strategies are distributed in proportion to the bikes mass, the selected r-strategies for all ingoing parts have also been measured in relation to the total weight of the electric two-wheeler, which can be seen in *Figure 12* below.

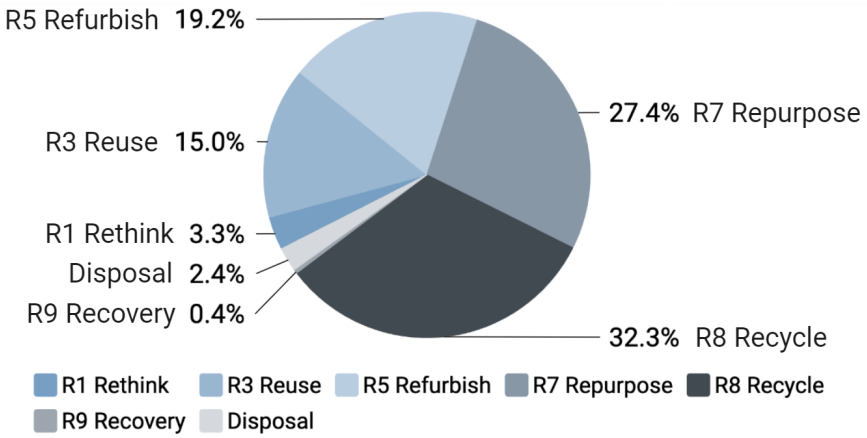


Figure 12: R-strategy Baseline, weight distribution.

As a result, the second life potential (strategy R3-R7) of the bike varies depending on which measurements are used. The disparity comes from the fact that mechanical fasteners and standard elements make up for a large amount of the components in the bill of materials but amounts for a small proportion of the bikes total weight. None of the mechanical fasteners or

standard elements such as screws, bolts, nuts or bearings are seen to have a second life potential. The product's second life potential can be seen in *Figure 13* below.

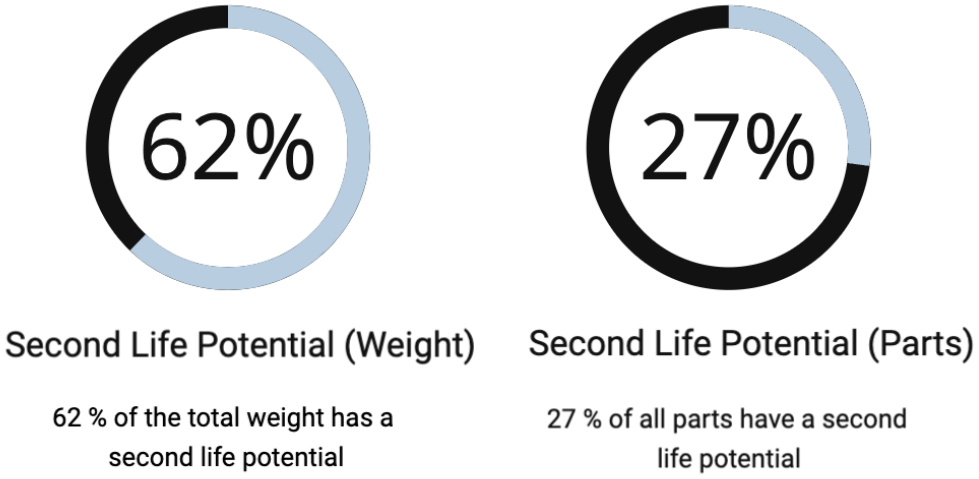


Figure 13: Second life potential of Makka:work.

5.1.4 R-strategy Verification

All r-strategies have been verified by using the verification method described in *section 4.1.4* to ensure that the correct hierarchical level was chosen for every part of the bike. It is included in the framework as it is seen as an important part of quality assuring the previous assessment, and that it should therefore not be forgotten or neglected. One example where the r-strategy was changed after posing the verification question was the footplates of the motorbike, where *R5 Refurbish* was first applied, but was later changed to *R1 Rethink* as it was realized that they were not designed in an optimal way from a durability perspective. The reason most parts could not receive higher r-strategies than they were first given was because most parts filled a crucial function and did not use unnecessary material amounts (i.e could not be given r-strategies *R0 Refuse* or *R2 Reduce*). Many of the parts given the r-strategy *R8 Recycle* could not climb the hierarchy as these parts were not seen to be in good enough condition to be reused, repaired or refurbished after a full product lifecycle. This could often be easily concluded by looking at how many spare part changes a part needed during one lifecycle. If a part needed to be changed during the first lifecycle, it naturally would not be suitable for reuse in a second lifecycle.

5.1.5 Practical Application

As CAKE has a functioning reverse logistics network within Sweden, as well as a second hand platform for refurbished motorbikes, r-strategies *R3 Reuse* and *R5 Refurbish* will be

applied by the service teams working within re:CAKE. Only one part of the motorbike received the r-strategy *R7 Repurpose* which is the battery, explaining why it amounts for 0.6% of the bike's parts but 27.4% of the bikes weight (See *Figure 11* and *Figure 12*). The most sustainable alternative would be to find a second life application for the battery once it is no longer suitable for use in a motorbike. However, as the batteries of the motorbikes are not as powerful as ones used in cars or trucks, the same second life applications for these vehicle batteries, such as energy storage or electricity grid stabilization, might not be as feasible or economically viable for energy companies (Interview Respondent 3; February 28, 2023). Therefore, there is a need to investigate other second life options for the batteries used in Makka:work. For strategy *R8 Recycle*, the main consideration will be whether the company should disassemble recyclable parts themselves or if they should be sent directly to the recycling centers unseparated. This will depend on the economic justification of manually separating the different components, as well as the environmental implications of not doing so. As seen in *Figure 10,* 84% of the motorbike Makka:work is estimated to be easy to disassemble, meaning most parts could be easily separated by the service teams and then be sent to recycling centers. Parts assigned with r-strategy *R9 Recovery* or *disposal* are either sent to incineration or landfill depending on whether there is justification for energy recovery or not. The strategy *R1 Rethink* is based on parts that are not manufactured or used in an ideal way, which calls for design changes or changes in manufacturing processes. How parts are to be rethought heavily depends on what the original problem of the part is seen to be, and the parts of Makka:work that were assigned this r-strategy will be described in further detail in the following section.

5.2 Design Improvements

The circular business model deliveries *cycling* and *extending* can be improved through design improvements. Changes in material choice and design are presented below. The presented changes have the objective of improving the second life potential and EoL-practices of the electric two-wheeler Makka:work. The motorbike and the parts with design improvement potential can be seen in *Figure 14* below.



Figure 14: Illustration of Makka:work and the parts where design improvements should be considered.

5.2.1 Material Choices

As mentioned in *4.2.1 Material Choices*, the main considerations when assessing the choice of materials are the use of *non-toxic substances, durable materials* and *recyclable materials*.

The motorbike Makka:work uses no toxic substances in its product design, and therefore no specific material choice improvements are suggested in the area. However, the batteries could have hazardous effects on their environment if not managed properly, emphasizing the notion of finding suitable second life or recycling options for the batteries.

When it comes to using more durable and recyclable materials, there are several components where this should be considered. The battery straps used to attach the battery to the bike get worn out relatively easily, causing a need to change them a needless number of times during the product life cycle. The same can be said about the saddle, where other, more durable materials should be investigated to decrease the amount of spare part changes needed during the product lifespan. The saddle also uses a combination of many different foams and plastics, making it infeasible for recycling. This in combination with the amount of wear and tear it sustains during its product life, makes it a component which should be investigated further. CAKE should either focus on extending the component lifespan by using more durable materials, or making it more recyclable by using materials which can be circulated. The rubber covers for the front and rear light are also components that are subjected to many spare part changes during a product life cycle. As rubber is not a recyclable material due to its polymer structure, a material change of the components should be considered from both a durability and a recyclability perspective. The parts mentioned are illustrated in *Figure 14*.

5.2.2 Design Choices

The five different aspects regarding CAKE's design choice improvements are presented in the following section.

Circular Product Design

Based on the r-strategy baseline of Makka:work, 68,9% of all product parts (See *Figure 11*) and 35,1% of the total mass of the bike (See *Figure 12*) fall under the categories *R8 Recycle*, *R9 Recovery* or *disposal* in the EoL-phase. To realize a closed loop of material and enhance a circular product design, CAKE should strive towards finding more solutions with r-strategies that fall under the hierarchical categorization *Reuse* (See r-strategy *R3-R7* in *Table 6*) by finding alternatives to current solutions that fall under categories lower down the hierarchy. As r-strategies in the hierarchical categorization *Reuse* promote multiple lifecycles for the components in the electric two-wheeler, increasing the number of components that fall under these categories in the EoL-phase would enhance a circular product design.

Design for Disassembly

Makka:work is already designed with DfD-principles in mind. However, there are still parts of the motorbike that could be improved from a disassembly perspective. The rear wheel is connected to the hub motor and the controller box in such a way that once one of these components needs to be disassembled, the entire rear wheel must be dismantled. Improvements should therefore be investigated for the rear wheel and its attached components to facilitate the EoL-management of the different parts of the assembly. Another module where DfD-improvements should be investigated is the seat assembly, where the entire seat assembly is replaced due to the saddle being worn out. If the seat assembly was more DfD-compatible, the seat post and the seat post clamp could be disassembled from the saddle as these components do not need to be replaced as often as the saddle does. The parts mentioned are illustrated in *Figure 14*.

Design for Modularity

By implementing modular systems between different product platforms, CAKE has an increased chance of finding second life possibilities at their own second hand platform re:CAKE. Further, the implementation of modularization and standardized interfaces allows CAKE to use smarter solutions from other bikes on the Makka:work. For instance, the battery straps become worn and need to be replaced many times during the lifecycle of Makka:work, but this is not the case on all of CAKE's models. By simply switching straps to the ones used in other models, since the straps already use a standardized interface, CAKE has an opportunity to improve Makka:work without redesigning any substructures of the electric two-wheeler.

Smart Product Design

Smarter product designs can improve current EoL-practices and enhance CE-principles. In the case of investigating the electric two-wheeler Makka:work, CAKE has an opportunity to avoid disposal (the least preferred EoL-option) by finding smarter solutions for the footplates. The current solution includes using tape on the plates to increase friction and prevent the

user's feet from slipping off the footplates. However, as the tape gets worn out, the footplates require maintenance to fulfill the same function over time. To decrease the need of maintenance and extend the part's life expectancy, CAKE should consider chequer plates as an alternate solution as it fulfills the same functional requirements. For similar reasons, an additional guideline is to redesign the cables of the displays as they sometimes break during the Swedish winter months, causing a need for them to be replaced more often than optimum (Interview Respondent 1, March 14, 2023). The brakes and the throttle should also be further investigated as these components have a lower lifetime than they likely could and need to be replaced during a product lifetime. The parts mentioned are illustrated in *Figure 14*.

Material & Component Reduction

Material & component reduction correspond to the r-strategies *R0 Refuse* and *R2 Reduce*. These r-strategies were not chosen for any part of the bike during its analysis, meaning that the bike is already stripped down of excessive material use and unnecessary design aesthetics. The motorbike is heavily focused on its functionality, light weight and simple design, and therefore no specific improvements were found in the area of material & component reduction.

5.3 Intensifying

As a significant amount of the electric two-wheeler Makka:work is being sold as a personal vehicle to B2B customers, CAKE should consider changes in their business models to enhance a circular economy and improve their EoL-management. To intensify the use phase of Makka:work, CAKE should move towards shared vehicle systems based on the following revenue models: *Subscription, Full Service Operating Lease, Rental,* and *Buyback* (See *Figure 15* below). While some of these models are already being implemented in the organization's current strategies, including these revenue models to a greater extent may favor EoL-management as CAKE receives more control of their vehicle. Since the implementation of these product-service system business models gives CAKE ownership of the vehicles, the organization also can decide what happens to their products in the EoL-stage.

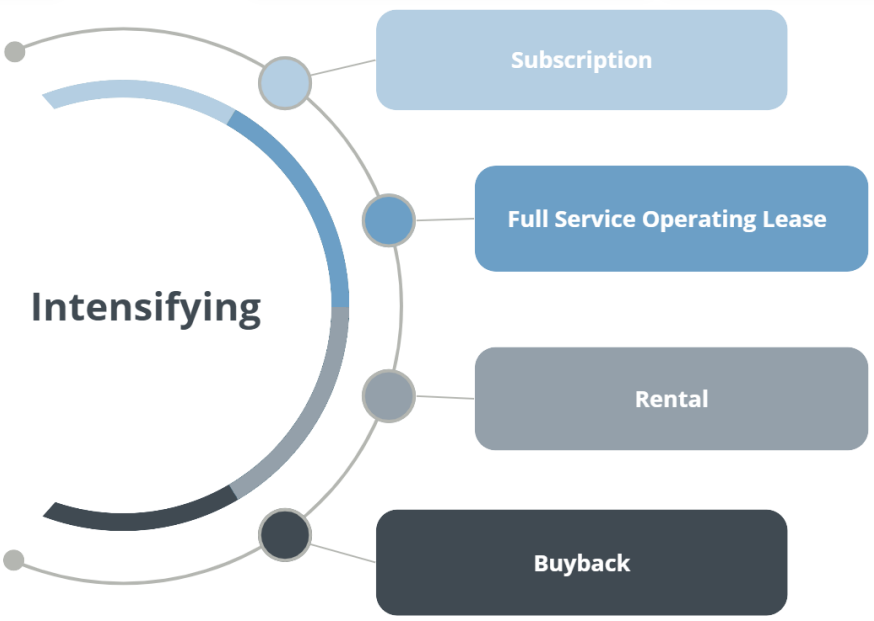


Figure 15: Revenue models that improve EoL-management

5.3.1 Subscription

Offering the value proposition as a subscription gives the customer access to multiple two-wheelers with no direct ownership of a specific vehicle. This also implies that a group of customers have access to the same amount of bikes, resulting in sharing the vehicles, intensifying the use-phase of the vehicle. By focusing on this model to a greater extent, CAKE has the opportunity to service some vehicles while others are being offered to the customers, truly focusing on product life extension and meeting customer expectations at the same time.

5.3.2 Full Service Operating Lease

A second option is offering the value proposition as a full service operating lease. While being offered to a single customer, CAKE has full ownership of the Makka:work and has the opportunity to control the way the customers interact with the vehicle during the use phase. Further, by including service as a part of the offering, CAKE has the opportunity of keeping the vehicle from obsolescence while controlling the final state of the electric two-wheeler after the life cycle.

5.3.3 Rental

Including rental options as a revenue model gives more customers an opportunity to travel in a sustainable manner. Despite being less preferred than full service operating leases due to less control of how the vehicle is used over time, renting the vehicles keeps CAKE as owners of the vehicle during the entire use phase of the product's lifecycle. While not being able to service the vehicle at any preferred time, keeping the ownership at CAKE can in turn

motivate the organization to make their products more durable and make service of their vehicles more efficient. Hence, including rental options can improve EoL-management.

5.3.4 Buyback

A buyback system of Makka:work would imply offering a fixed price for CAKE to repurchase the vehicle from the customer after its first lifecycle. The price can depend on several factors such as distance traveled or time after original purchase. This system implies that the customer has full ownership of the vehicle during the use phase of the bike. Nevertheless, by assuring the customer that CAKE is willing to repurchase their vehicles under certain conditions, some users might change their behavior to keep the initial value of the bike to their greatest extent. Further, a buyback system simplifies reverse logistics and promotes second life possibilities since the buyback requirements are based on valuable conditions of the bike, i.e., conditions for second life possibilities. In turn, creating a buyback system can simplify EoL-management.

6. Discussion

In this chapter, discussion points from various parts of the project work are highlighted. The chapter is divided into the main sections of the thesis work, namely *Purpose & Delimitations, Frame of Reference, Method, Development of Framework* and *Framework Implementation*.

6.1 Purpose & Delimitations

The EoL-evaluation of the motorbike Makka:work is based on the product having a technical lifespan of 20 years or a travel distance of 600 000 km. If the technical lifespan was estimated to be either longer or shorter, the assigned r-strategies would most likely be distributed differently. The results of the r-strategy baseline are therefore deeply connected to the assigned life length of the two-wheeler.

In addition, the thesis work has been mainly focused on the environmental sustainability perspective, neglecting the aspects of social and economic sustainability. For the end-of-life practices to be fully sustainable, all three aspects should be considered. By including social and economic sustainability, aspects such as second life applications of Li-ion batteries and inclusions of other materials in the product design would have been analyzed from perspectives such as human health and economic viability.

6.2 Frame of Reference

The EoL-management of lithium-ion batteries has been researched in this project but has not been analyzed in great detail for CAKE. For this thesis, it is specified that the batteries have been excluded in *1.3 Delimitations* due to CAKE already having a team investigating the subject. The reason for still including it in the Frame of Reference is that it is an important aspect for all electric vehicle manufacturers to consider. Even though it has not been researched in this thesis for CAKE, it should be emphasized that other companies should not neglect investigating second-life applications for LIBs. While the case company has a separate team investigating the area, it is not guaranteed that all organizations implementing the thesis framework do. Therefore, despite not having detailed research of LIBs in this project, organizations that develop and manufacture electric vehicles should not neglect finding sustainable EoL-practices for LIBs. Further, while selecting the r-strategy *R7 Repurpose* (See *Table 1*) as an appropriate EoL-option for CAKE's batteries, no specific second-life application was stated in this project.

6.3 Method

While the respondents that validated key aspects of topics included in this thesis can be seen as reliable, interviews with more respondents confirming the same information could help validate the primary data received in this thesis. During the project, semi-structured interviews were held with the respondents, each covering an aspect that is crucial for improving EoL-management and enhancing a circular economy. However, while some

experts stated similar matters in these fields, further actions could be taken to cover the same topics with different respondents. Although twelve respondents were interviewed in this thesis (See *Table 4*), an additional ten respondents were contacted but were not available during the timeframe of the project. By interviewing more respondents with the same fields of expertise, further validations on the accumulated information could be made if the experts answered in similar ways.

Additionally, the CAKE assembly site was visited only once during the project execution. During this visit, there was no time to investigate the potential assembly and disassembly difficulties of the product Makka:work. By visiting the CAKE assembly site more frequently, more extensive investigations of the disassembly levels of the different product parts could have been made, making the disassembly evaluation more detailed and verifiable.

6.4 Development of Framework

In the framework of this thesis, the r-strategies *R0 Refuse*, and *R1 Rethink* have been interpreted differently than the definition accumulated from previous studies (See *Table 1* and *Table 6*). Potting et al. (2017) describes the r-strategy *Refuse* as *"making a product redundant by finding alternate solutions that provide the same function"*. However, since this project involves analyzing all product parts of Makka:work rather than identifying an r-strategy for the bike as a whole, abandoning the functions of the bike or creating a radically different product is not in the scope of the project. Further, the r-strategy *Rethink* is defined as a strategy to make product use more intensified, also focusing on a product as a whole rather than analyzing each product component (ibid). Consequently, these two r-strategies have been interpreted differently in this thesis (See *Table 6*). However, neither strategy described in the frame of reference has been neglected due to being crucial for improving EoL-management. Intensifying the use of the electric two-wheeler, the definition of *Rethink* by Potting et al. (2017), has been included in a separate section of the framework (See *Figure 6*) and finding alternate solutions that provide the same functions, *Refuse,* has been adapted to be applicable on a component level (See *Table 6*).

Further, even though the definition of life extension is relatively similar in the literature used for the frame of reference, there is a variety in how it is applied in different models. Potting et al. (2017) define it as the r-strategies R3-R7 (Reuse, repair, refurbish, remanufacture, repurpose), while Geissdoerfer et al. (2020) make a distinction between cycling and life extension, where cycling corresponds to the r-strategies reuse, remanufacturing, refurbishing and recycling, and life extension is instead defined as "long lasting design, marketing, maintenance and repair". In this thesis, the definition is more in line with the definition of Potting et al. (2017), as life extension is more applied on a component level rather than a product level. Therefore, life extension has not covered aspects such as emotional/style product obsolescence, upgrading and marketing.

The r-strategies can be prioritized differently when further investigating more markets. Since most organizations, including the case company, have various capabilities in different

regional markets, the EoL-possibilities might alter between markets. These capabilities may include tangible and intangible resources, relations and capabilities with immediate stakeholders, infrastructure, and differences between cultures. While CAKE has several service stations located in Sweden, this is not the case for all markets where CAKE's vehicles are used. Since the r-strategies *Repair*, *Refurbish*, and *Remanufacture* heavily rely on service stations who can repair damaged vehicles, the objective of prolonging the lifetime of the electric two-wheelers is complicated in markets without service stations. The same can be said about the RL-network, where the reverse logistics used on the Swedish market also rely on service stations and their supporting infrastructure. The prospect of using a buyback-system is dependent on having a second hand platform where refurbished motorbikes can be sold, as is the case with re:CAKE on the Swedish market. Therefore, the same RL-capabilities may not be applicable on other markets. Further, since the recycling rate and the recycling process of different materials may alter between different markets, the possibilities of recycling and its benefits also differ. As acknowledged by Soo et al. (2017), the demand of different materials affects the recycling rate and recycling efficiency of the material, which affects how ELVs are recycled in different markets. For instance, the recycling efficiency of iron in Australia is relatively high in comparison to other metals due to the demand to provide stock for their affiliated steel mill company (ibid). Therefore, cultural differences between markets make the priority of recycling different materials differ. Additionally, while some recycling processes alter due to demands of different materials, the recycling efficiency of different materials can also alter between developed, underdeveloped, and developing countries. Even though recycling is highly prioritized in developed countries, this is not the case on a global level. As a result, the evaluation of applicable and suitable r-strategies can differ when investigating more markets.

Design for Recycling can improve the current EoL-practices of vehicles. This design strategy glorifies recycling as an EoL-strategy. However, as described by the Waste Framework Directive (European Commission, n.d.-a), recycling is one of the least preferred options to realize a circular economy since an excessive amount of energy is required to separate material, recycle material, and manufacture new products from the recycled goods. Moreover, if recycling is considered in a market without renewable energy sources, the recycling process will have a negative impact on the environment, especially in comparison to higher EoL-alternatives in the waste waste management hierarchy model. Thus, when implementing DfR as a design strategy to improve the sustainable practices of an organization, more sustainable design choices are neglected. Nevertheless, under circumstances where recycling is the best possible EoL-option, improving ways to recycle is beneficial from a social, environmental, and economic perspective. Therefore, considering and combining different design strategies can ultimately result in optimizing an organization's EoL-strategies.

The method used to evaluate the level of disassembly of all product parts uses three evaluation categories, namely *Information*, *Tools*, and *Accessibility*. The reason for including these three categories only was to give a fundamental understanding of the level of disassembly in the timeline of the project while minimizing the need for information from internal staff at CAKE. However, multiple aspects discovered from both primary and

secondary data were excluded when creating the method used in this project. Aspects like *time to disassemble* and *positioning*, which could result in a more complete disassembly evaluation (See Figure 3), would require further tests at the assembly site for Makka:work. Due to lack of time and lack of access to the assembly site, these aspects were excluded in the final method for evaluating the level of disassembly. However, an assembly instruction manual was used to get insights into how service personnel operate when disassembling different product parts. Further, in contrast to an evaluation method presented by Respondent 7, the evaluation method used in this project does not make a distinction between the information needed for certified service personnel and consumers to dismantle product parts from the Makka:work. By excluding customer inputs, the level of disassembly is directed towards service center personnel only. Hence, while consumers can dismantle parts themselves, the disassembly evaluation method used in this thesis does not apply for the replacement of parts by consumers. The aspect of evaluating disassembly from a consumer perspective was neglected due to lack of B2B-customer inputs when conducting the assessment. Lastly, in comparison to the methods received from primary data and secondary data, the evaluation method used in this thesis only uses three numeric values for each evaluation category, resulting in three different levels of disassembly. By including more numeric values, further differences between the disassembly of different parts could be discovered. However, due to the time limitation of the project, three numeric values were selected to give the case company an overview of their products' level of disassembly.

6.5 Implementation of Framework

As Makka:work has been analyzed on a component level rather than a product level, some r-strategies have been neglected as they were not seen as practically feasible. As shown in the evaluation of suitable r-strategies for Makka:work (see *Figure 11 and Figure 12*), *Refuse, Reduce, Repair,* and *Remanufacture* were not seen as preferred r–strategies for any of the product parts based on the organizational capabilities and the product properties. *Refuse* and *Reduce* have been neglected since all components have been evaluated as necessary for providing functional value to the electric two-wheeler. However, materials used in some product components could possibly be reduced if analyzed in greater detail. Nevertheless, due to time limitations and the prioritizing of other parts of the project scope, this aspect has been neglected. Further, as CAKE outsources the production of a majority of the product components, *Remanufacture* has not been chosen as a feasible r-strategy since it is not possible without including stakeholders outsourced from CAKE. For the same reason, *Repair* of the product components has not been chosen as a feasible r-strategy either.

Regarding the practical application of the evaluation method, the ingoing parts of Makka:work were weighed using two weight scales of different sizes at CAKE's assembly site, where the scale of choice depended on the size of the product parts. However, the larger scale had a margin of error of ± 50 grams, which resulted in an uncertain estimate of the total mass of Makka:work. Further, when also assuming the weight of 205 mechanical fasteners to be of 5 grams each, the total mass of the bike was summed up to 83 886 g, exceeding the mass of 82 000 g given in the product's technical description by CAKE with 1 886 g (CAKE,

n.d.). Accordingly, the margin of error of the weight scale as well as the weight estimation of the mechanical fasteners are the biggest reasons for the disparity between the weight in the product's technical description and the summed-up weight in this project work. Further, all mechanical fasteners were assigned with *Recycle* as the most suitable r-strategy. Therefore, assuming the weight of the mechanical fasteners and having a relatively high margin of error when weighing the larger product component can have resulted in an inaccurate distribution of selected r-strategies in comparison to the total mass of the electric two-wheeler.

As previously established by Potting et al. (2017), the hierarchical category *R3 Reuse* is generally preferable over *R8 Recycling*. However, it is not a realistic alternative for all product components due to wear and tear, resulting in quality and safety issues. This is the case for the mechanical fasteners used in Makka:work. Respondent 1 and Respondent 4 both stated that mechanical fasteners should not be reused due to the resources required to dismantle them, as well as quality deficiencies in the components. Therefore, when looking at the r-strategies distributed to the ingoing parts of Makka:work (see *Figure 11 and Figure 12*), the results might look skewed as all mechanical fasteners have received the r-strategy *R8 Recycling,* making it seem like the bike has less second life potential than it possibly could have, when in reality there are no realistic prospects of reusing them. The use of mechanical fasteners is also necessary from the perspective of disassembly. Adhesive bonding would reduce the need for mechanical fasteners but would decrease the disassembly levels making the product more difficult to repair, refurbish and reuse. Since the mechanical fasteners are a relatively small proportion of the total mass of the electric two-wheeler, mechanical fasteners are preferred over adhesive bondings as they increase level of disassemblability.

Evaluating all product parts resulted in general improvements in both design and business models. However, analyzing fewer components could have resulted in a more specific set of guidelines for a select few. Several areas of improvement have been highlighted after applying the framework on the product Makka:work. However, evaluating every single product component requires a lot of time and effort, which has limited the possibilities to investigate exactly how the product design and the business model could be improved. As a result, most guidelines highlight where certain changes are needed, but not exactly how these changes should be implemented. One exception is the specific recommendation of replacing the current tape on the foot plates with chequers plates to maintain the functional value and extend the life expectancy of the product component, which was given to exemplify how the framework can be used for product improvements.

The prospect of using a sharing model where customers rent the motorbikes by the hour or for short periods of time has been excluded from the intensifying part of Makka:work's business model. The argument is that since Makka:work is a motorbike for B2B-customers, very few customers would have a need for renting the bike by the hour as they often need the bikes over longer periods of time. It would also require different B2B-customers to compete for the same product supply, which would cause uncertainties over the access of motorbikes for every customer. Therefore, intensifying the use of Makka:work has focused on sharing

solutions where the customers have ownership of the motorbike under longer arrangement periods.

A life cycle assessment (LCA) of Makka:work would help prioritize the set of actions recommended to CAKE in the order that helps the organization decrease their environmental footprint from most to least. When creating or improving a product with a goal of minimizing the environmental impact, LCA:s can be used to shape the process of which products are designed (Ehrenfeld, 1997). However, an LCA of Makka:work was conducted at the same time as this thesis work was performed. This meant the results from the LCA could not be used in this thesis work as it was not completed in time for it to be included in this project work. As a result, there are uncertainties of the environmental impact from each product component resulting in difficulties in prioritizing the design changes to decrease the environmental impact of the product. Further, LCA values could also change the direction of the thesis scope as additional crucial aspects could have been discovered when analyzing the product life cycle.

7. Conclusions

In this chapter, the concept and usage of circular business models have been investigated to help make electric two-wheelers more sustainable in the EoL-phase. Further, all conclusions are intended to answer the main objectives described in section *1.2. Purpose*.

A framework has been developed to cover the key aspects of end-of-life management. The included aspects were chosen based on what was seen to be crucial from scientific research and expert interviews. The framework can be seen as a baseline for how EoL-practices can be initiated, but when the foundations for each included aspect has been investigated, more methods and tools should be included to ensure that the end-of-life practices are practically and economically viable. Regarding the contribution to the field of research, the developed framework has been adapted to be applied on a component level rather than on a product level to decrease the environmental burden of a product on multiple levels. The accumulation of crucial aspects of EoL-management developed into a comprehensive framework can therefore be seen as the contribution to the research area.

The framework was implemented on the electric two-wheeler Makka:work developed by Cake 0 Emission. To decrease the environmental impact during the EoL-phase of Makka:work and promote a closed loop of material, guidelines are set for CAKE to make changes in design and material of choice for several product components. Further, additional changes in value proposition and value capture are recommended to help change the behavior of users and improve current EoL-practices. However, while a set of improvements have been discovered using this framework, future work is required to identify feasible solutions in these areas. Nevertheless, by developing a framework with key aspects of EoL-management and implementing it on the case company, the main objectives of the study have been completed.

8. Future Work

In this chapter, future work regarding improvements of the developed framework and its practical application are suggested. Additionally, to include various markets and realize sustainable EoL-practices on a global level, future work is recommended on investigating different regional markets.

To apply the framework in regional markets outside of Sweden, further research is required regarding reverse logistics and how different r-strategies can be implemented. Since the case company has various capabilities outside of Sweden, the EoL-possibilities will alter between different regional markets. The level of available service locations, the electricity mix, culture, and infrastructure will affect the suitable EoL-options for the ingoing parts of the electric two-wheeler.

To improve the evaluation of the bike and identify further differences on the level of disassembly for all product parts, the method for evaluating the level of disassembly should be investigated further. By including aspects like time to disassemble and the force required to dismantle parts from the bike, a more complete understanding of the level of disassembly can be identified. Further, collecting data regarding time of disassembly and force required when dismantling product parts is required to analyze the cost of implementing r-strategies in the hierarchical categorization *Reuse* (See *Table 6*). To include these aspects in the evaluation model, several tests with the service department or the personnel at the assembly site need to be performed to receive objective measurements.

Several areas of improvements in material of choice and design have been identified by the practical application of the framework to move the case company towards more sustainable EoL-practices. However, specific solutions in these areas are not included in the scope of this project. Hence, for CAKE to improve the electric two-wheeler Makka:work in practice, future work is required to develop specific sustainable solutions in these areas. Moreover, further research into the economic viability of each revenue model should be able to determine which shared vehicle solutions are preferred from both an environmental and economic perspective.

The economic and social considerations of sustainability should be further researched to cover all three aspects of sustainability. Different cost estimation methods, such as CBA or life cycle cost (LCC) analysis, should be considered to investigate the economic viability of different EoL-strategies and circular business model deliveries. Social considerations should also be investigated to make sure that parts of the supply chain and business model deliveries do not conflict with important human health, justice and equality perspectives. As the LCA of Makka:work was not completed in time for it to be included in this project, an analysis of design changes in regards to the environmental impact of different product parts should be considered. Without knowing the environmental impacts of the different product parts, the proposed design improvements cannot be prioritized from most to least critical from an

environmental perspective. Thereby, the importance of the different improvement proposals would be easier to determine with the inclusion of the LCA.

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